

IMPORTING LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

```
df=pd.read_csv(r"C:\Users\91798\Desktop\Car Brand.csv")
```

```
df
```

Engine HP	Make	Model	Year	Engine Fuel Type
0	BMW	1 Series M	2011	premium unleaded (required)
335.0				
1	BMW	1 Series	2011	premium unleaded (required)
300.0				
2	BMW	1 Series	2011	premium unleaded (required)
300.0				
3	BMW	1 Series	2011	premium unleaded (required)
230.0				
4	BMW	1 Series	2011	premium unleaded (required)
230.0				
...	...	...	...	...
...				
11909	Acura	ZDX	2012	premium unleaded (required)
300.0				
11910	Acura	ZDX	2012	premium unleaded (required)
300.0				
11911	Acura	ZDX	2012	premium unleaded (required)
300.0				
11912	Acura	ZDX	2013	premium unleaded (recommended)
300.0				
11913	Lincoln	Zephyr	2006	regular unleaded
221.0				

Engine Cylinders	Transmission Type	Driven_Wheels	Number of Doors
6.0	MANUAL	rear wheel drive	0
2.0			
6.0	MANUAL	rear wheel drive	1
2.0			
6.0	MANUAL	rear wheel drive	2
2.0			
6.0	MANUAL	rear wheel drive	3
2.0			
6.0	MANUAL	rear wheel drive	4
2.0			

...	...	...	...
11909	6.0	AUTOMATIC	all wheel drive
4.0			
11910	6.0	AUTOMATIC	all wheel drive
4.0			
11911	6.0	AUTOMATIC	all wheel drive
4.0			
11912	6.0	AUTOMATIC	all wheel drive
4.0			
11913	6.0	AUTOMATIC	front wheel drive
4.0			

Style \	Market Category	Vehicle Size	Vehicle
0	Factory Tuner,Luxury,High-Performance	Compact	
Coupe			
1	Luxury,Performance	Compact	
Convertible			
2	Luxury,High-Performance	Compact	
Coupe			
3	Luxury,Performance	Compact	
Coupe			
4	Luxury	Compact	
Convertible			
...	...	...	.

11909	Crossover,Hatchback,Luxury	Midsize	4dr
Hatchback			
11910	Crossover,Hatchback,Luxury	Midsize	4dr
Hatchback			
11911	Crossover,Hatchback,Luxury	Midsize	4dr
Hatchback			
11912	Crossover,Hatchback,Luxury	Midsize	4dr
Hatchback			
11913	Luxury	Midsize	
Sedan			

	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500
...	...	...	...	...
11909	23	16	204	46120
11910	23	16	204	56670
11911	23	16	204	50620
11912	23	16	204	50920

```
11913          26          17          61  28995
[11914 rows x 16 columns]
```

DATA DESCRIPTION

```
df.shape          ## TO FIND THE NUMBER OF ROWS AND COLUMNS IN A DATA
(11914, 16)
```

```
df.columns        ## TO FIND THE COLUMN NAMES OF THE DATA
```

```
Index(['Make', 'Model', 'Year', 'Engine Fuel Type', 'Engine HP',
      'Engine Cylinders', 'Transmission Type', 'Driven_Wheels',
      'Number of Doors', 'Market Category', 'Vehicle Size', 'Vehicle
      Style',
      'highway MPG', 'city mpg', 'Popularity', 'MSRP'],
      dtype='object')
```

```
df.dtypes         ## TO FIND THE TYPE OF THE DATA
```

```
Make          object
Model         object
Year          int64
Engine Fuel Type  object
Engine HP     float64
Engine Cylinders float64
Transmission Type  object
Driven_Wheels  object
Number of Doors  float64
Market Category  object
Vehicle Size    object
Vehicle Style   object
highway MPG     int64
city mpg        int64
Popularity      int64
MSRP            int64
dtype: object
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11914 entries, 0 to 11913
Data columns (total 16 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Make                 11914 non-null  object
 1   Model                11914 non-null  object
 2   Year                 11914 non-null  int64
 3   Engine Fuel Type     11911 non-null  object
 4   Engine HP            11845 non-null  float64
```

```

5   Engine Cylinders    11884 non-null float64
6   Transmission Type   11914 non-null object
7   Driven_Wheels       11914 non-null object
8   Number of Doors     11908 non-null float64
9   Market Category     8172 non-null object
10  Vehicle Size        11914 non-null object
11  Vehicle Style       11914 non-null object
12  highway MPG         11914 non-null int64
13  city mpg            11914 non-null int64
14  Popularity          11914 non-null int64
15  MSRP                11914 non-null int64

```

dtypes: float64(3), int64(5), object(8)

memory usage: 1.5+ MB

df.describe()

TO GET COMPLETE

INFO

	Year	Engine HP	Engine Cylinders	Number of Doors \
count	11914.000000	11845.00000	11884.000000	11908.000000
mean	2010.384338	249.38607	5.628829	3.436093
std	7.579740	109.19187	1.780559	0.881315
min	1990.000000	55.00000	0.000000	2.000000
25%	2007.000000	170.00000	4.000000	2.000000
50%	2015.000000	227.00000	6.000000	4.000000
75%	2016.000000	300.00000	6.000000	4.000000
max	2017.000000	1001.00000	16.000000	4.000000

	highway MPG	city mpg	Popularity	MSRP
count	11914.000000	11914.000000	11914.000000	1.191400e+04
mean	26.637485	19.733255	1554.911197	4.059474e+04
std	8.863001	8.987798	1441.855347	6.010910e+04
min	12.000000	7.000000	2.000000	2.000000e+03
25%	22.000000	16.000000	549.000000	2.100000e+04
50%	26.000000	18.000000	1385.000000	2.999500e+04
75%	30.000000	22.000000	2009.000000	4.223125e+04
max	354.000000	137.000000	5657.000000	2.065902e+06

DATA CLEANING

##Missing values

df.isna().sum()

TO FIND MISSING VALUES

```

Make          0
Model         0
Year          0
Engine Fuel Type  3
Engine HP      69
Engine Cylinders 30
Transmission Type  0
Driven_Wheels  0

```

```

Number of Doors      6
Market Category      3742
Vehicle Size         0
Vehicle Style        0
highway MPG         0
city mpg             0
Popularity           0
MSRP                 0
dtype: int64

```

```

df.rename(columns={'Engine Fuel Type':'Fuel Type',
                  'Engine Cylinders':'Cylinders',
                  'Vehicle Size':' size',
                  'Engine HP':'Horse Power'},inplace=True)

```

```
df
```

	Make	Model	Year	Fuel Type
Horse Power \				
0	BMW	1 Series M	2011	premium unleaded (required)
335.0				
1	BMW	1 Series	2011	premium unleaded (required)
300.0				
2	BMW	1 Series	2011	premium unleaded (required)
300.0				
3	BMW	1 Series	2011	premium unleaded (required)
230.0				
4	BMW	1 Series	2011	premium unleaded (required)
230.0				
...	...	...	...	...
...				
11909	Acura	ZDX	2012	premium unleaded (required)
300.0				
11910	Acura	ZDX	2012	premium unleaded (required)
300.0				
11911	Acura	ZDX	2012	premium unleaded (required)
300.0				
11912	Acura	ZDX	2013	premium unleaded (recommended)
300.0				
11913	Lincoln	Zephyr	2006	regular unleaded
221.0				

	Cylinders	Transmission Type	Driven_Wheels	Number of Doors
\				
0	6.0	MANUAL	rear wheel drive	2.0
1	6.0	MANUAL	rear wheel drive	2.0
2	6.0	MANUAL	rear wheel drive	2.0

3	6.0	MANUAL	rear wheel drive	2.0
4	6.0	MANUAL	rear wheel drive	2.0
...	...	...	...	...
11909	6.0	AUTOMATIC	all wheel drive	4.0
11910	6.0	AUTOMATIC	all wheel drive	4.0
11911	6.0	AUTOMATIC	all wheel drive	4.0
11912	6.0	AUTOMATIC	all wheel drive	4.0
11913	6.0	AUTOMATIC	front wheel drive	4.0

Style \	Market Category	size	Vehicle
0	Factory Tuner,Luxury,High-Performance	Compact	Coupe
1	Luxury,Performance	Compact	Convertible
2	Luxury,High-Performance	Compact	Coupe
3	Luxury,Performance	Compact	Coupe
4	Luxury	Compact	Convertible
...	...	...	...
11909	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11910	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11911	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11912	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11913	Luxury	Midsize	Sedan

	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500
...	...	...	...	...
11909	23	16	204	46120

11910	23	16	204	56670
11911	23	16	204	50620
11912	23	16	204	50920
11913	26	17	61	28995

[11914 rows x 16 columns]

```
df['Fuel Type'].unique()
```

```
array(['premium unleaded (required)', 'regular unleaded',
      'premium unleaded (recommended)', 'flex-fuel (unleaded/E85)',
      'diesel', 'electric',
      'flex-fuel (premium unleaded recommended/E85)', 'natural gas',
      'flex-fuel (premium unleaded required/E85)',
      'flex-fuel (unleaded/natural gas)', nan], dtype=object)
```

```
df[df['Fuel Type'].isna()]
```

	Make	Model	Year	Fuel Type	Horse Power	Cylinders \
11321	Suzuki	Verona	2004	NaN	155.0	6.0
11322	Suzuki	Verona	2004	NaN	155.0	6.0
11323	Suzuki	Verona	2004	NaN	155.0	6.0

	Transmission Type	Driven_Wheels	Number of Doors	Market Category \
11321	AUTOMATIC	front wheel drive	4.0	NaN
11322	AUTOMATIC	front wheel drive	4.0	NaN
11323	AUTOMATIC	front wheel drive	4.0	NaN

	size	Vehicle Style	highway MPG	city mpg	Popularity	MSRP
11321	Midsize	Sedan	25	17	481	17199
11322	Midsize	Sedan	25	17	481	20199
11323	Midsize	Sedan	25	17	481	18499

```
df[df['Make']=='Suzuki']['Fuel Type'].unique()
```

```
array(['regular unleaded', nan], dtype=object)
```

```
df['Fuel Type'].fillna('regular unleaded',inplace=True)
```

```
df.isna().sum()
```

Make	0
Model	0
Year	0

```
Fuel Type          0
Horse Power        69
Cylinders          30
Transmission Type  0
Driven_Wheels      0
Number of Doors    6
Market Category    3742
size              0
Vehicle Style      0
highway MPG        0
city mpg           0
Popularity         0
MSRP               0
dtype: int64
```

```
df[df['Horse Power'].isna()][['Transmission Type']].unique()
array(['DIRECT_DRIVE', 'AUTOMATIC'], dtype=object)
```

```
df.isna().sum()
```

```
Make          0
Model         0
Year          0
Fuel Type     0
Horse Power   69
Cylinders     30
Transmission Type  0
Driven_Wheels 0
Number of Doors 6
Market Category 3742
size          0
Vehicle Style 0
highway MPG    0
city mpg       0
Popularity     0
MSRP           0
dtype: int64
```

```
df.isna().sum()
```

```
Make          0
Model         0
Year          0
Fuel Type     0
Horse Power   69
Cylinders     30
Transmission Type  0
Driven_Wheels 0
Number of Doors 6
Market Category 3742
```



```
size          0
Vehicle Style 0
highway MPG   0
city mpg      0
Popularity    0
MSRP          0
dtype: int64
```

```
df['Cylinders'].unique()
```

```
array([ 6.,  4.,  5.,  8., 12.,  0., nan, 10.,  3., 16.])
```

```
df['Cylinders'].fillna(df['Cylinders'].median(),inplace=True)
```

```
df[df['Number of Doors'].isnull()]
```

	Make	Model	Year	Fuel Type	Horse Power
4666	Ferrari	FF	2013	premium unleaded (required)	651.0
6930	Tesla	Model S	2016	electric	NaN
6931	Tesla	Model S	2016	electric	NaN
6932	Tesla	Model S	2016	electric	NaN
6933	Tesla	Model S	2016	electric	NaN
6934	Tesla	Model S	2016	electric	NaN

	Cylinders	Transmission Type	Driven_Wheels	Number of Doors
4666	12.0	AUTOMATED_MANUAL	all wheel drive	NaN
6930	0.0	DIRECT_DRIVE	all wheel drive	NaN
6931	0.0	DIRECT_DRIVE	all wheel drive	NaN
6932	0.0	DIRECT_DRIVE	all wheel drive	NaN
6933	0.0	DIRECT_DRIVE	rear wheel drive	NaN
6934	0.0	DIRECT_DRIVE	all wheel drive	NaN

	Market Category	size	Vehicle Style	highway MPG	city mpg
4666	Exotic,High-Performance	Large	Coupe	16	11
6930	Exotic,Performance	Large	Sedan	105	102

693198	Exotic,Performance	Large	Sedan	101
693292	Exotic,High-Performance	Large	Sedan	105
693397	Exotic,Performance	Large	Sedan	100
6934101	Exotic,Performance	Large	Sedan	107

	Popularity	MSRP
4666	2774	295000
6930	1391	79500
6931	1391	66000
6932	1391	134500
6933	1391	74500
6934	1391	71000

```
df['Number of Doors'] = df.groupby(by = 'Make')['Number of Doors'].transform(lambda x: x.fillna(x.mode()[0]))
```

```
df
```

	Make	Model	Year	Fuel Type
Horse Power \				
0	BMW	1 Series M	2011	premium unleaded (required)
335.0				
1	BMW	1 Series	2011	premium unleaded (required)
300.0				
2	BMW	1 Series	2011	premium unleaded (required)
300.0				
3	BMW	1 Series	2011	premium unleaded (required)
230.0				
4	BMW	1 Series	2011	premium unleaded (required)
230.0				
...	...	...	...	...
...				
11909	Acura	ZDX	2012	premium unleaded (required)
300.0				
11910	Acura	ZDX	2012	premium unleaded (required)
300.0				
11911	Acura	ZDX	2012	premium unleaded (required)
300.0				
11912	Acura	ZDX	2013	premium unleaded (recommended)
300.0				
11913	Lincoln	Zephyr	2006	regular unleaded
221.0				

	Cylinders	Transmission	Type	Driven_Wheels	Number of Doors
\					
0	6.0	MANUAL	rear wheel drive		2.0

1	6.0	MANUAL	rear wheel drive	2.0
2	6.0	MANUAL	rear wheel drive	2.0
3	6.0	MANUAL	rear wheel drive	2.0
4	6.0	MANUAL	rear wheel drive	2.0
...	...	...	...	...
11909	6.0	AUTOMATIC	all wheel drive	4.0
11910	6.0	AUTOMATIC	all wheel drive	4.0
11911	6.0	AUTOMATIC	all wheel drive	4.0
11912	6.0	AUTOMATIC	all wheel drive	4.0
11913	6.0	AUTOMATIC	front wheel drive	4.0

Style	Market Category	size	Vehicle
0	Factory Tuner,Luxury,High-Performance	Compact	Coupe
1	Luxury,Performance	Compact	Convertible
2	Luxury,High-Performance	Compact	Coupe
3	Luxury,Performance	Compact	Coupe
4	Luxury	Compact	Convertible
...	...	...	...
11909	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11910	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11911	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11912	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11913	Luxury	Midsize	Sedan

	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350

3	28	18	3916	29450
4	28	18	3916	34500
...	...	...	...	...
11909	23	16	204	46120
11910	23	16	204	56670
11911	23	16	204	50620
11912	23	16	204	50920
11913	26	17	61	28995

[11914 rows x 16 columns]

```
df.isnull().sum()
```

Make	0
Model	0
Year	0
Fuel Type	0
Horse Power	69
Cylinders	0
Transmission Type	0
Driven_Wheels	0
Number of Doors	0
Market Category	3742
size	0
Vehicle Style	0
highway MPG	0
city mpg	0
Popularity	0
MSRP	0

dtype: int64

```
df['Market Category'].isna()
```

0	False
1	False
2	False
3	False
4	False
...	...
11909	False
11910	False
11911	False
11912	False
11913	False

Name: Market Category, Length: 11914, dtype: bool

```
df['Market Category'].fillna(df['Market Category'].mode()[0],inplace=True)
```

```
df.isna().sum()
```

```
Make          0
Model         0
Year          0
Fuel Type     0
Horse Power   69
Cylinders     0
Transmission Type 0
Driven_Wheels 0
Number of Doors 0
Market Category 0
size         0
Vehicle Style 0
highway MPG   0
city mpg      0
Popularity    0
MSRP          0
dtype: int64
```

```
df.duplicated().sum()
```

```
715
```

```
df[df.duplicated()]
```

	Make	Model	Year	Fuel Type	Horse
Power \					
14	BMW	1 Series	2013	premium unleaded (required)	
230.0					
18	Audi	100	1992	regular unleaded	
172.0					
20	Audi	100	1992	regular unleaded	
172.0					
24	Audi	100	1993	regular unleaded	
172.0					
25	Audi	100	1993	regular unleaded	
172.0					
...	...	...	...	...	..
.					
11481	Suzuki	X-90	1998	regular unleaded	
95.0					
11603	Volvo	XC60	2017	regular unleaded	
302.0					
11604	Volvo	XC60	2017	regular unleaded	
240.0					
11708	Suzuki	XL7	2008	regular unleaded	
252.0					
11717	Suzuki	XL7	2008	regular unleaded	
252.0					

Cylinders	Transmission Type	Driven_Wheels	Number of Doors
-----------	-------------------	---------------	-----------------

\	14	6.0	MANUAL	rear wheel drive	2.0
	18	6.0	MANUAL	front wheel drive	4.0
	20	6.0	MANUAL	front wheel drive	4.0
	24	6.0	MANUAL	front wheel drive	4.0
	25	6.0	MANUAL	front wheel drive	4.0
	...	...	...	...	...
	11481	4.0	MANUAL	four wheel drive	2.0
	11603	4.0	AUTOMATIC	all wheel drive	4.0
	11604	4.0	AUTOMATIC	front wheel drive	4.0
	11708	6.0	AUTOMATIC	all wheel drive	4.0
	11717	6.0	AUTOMATIC	front wheel drive	4.0
	Market Category		size	Vehicle Style	highway
MPG \	14	Luxury,Performance	Compact	Coupe	
	28				
	18	Luxury	Midsize	Sedan	
	24				
	20	Luxury	Midsize	Sedan	
	24				
	24	Luxury	Midsize	Sedan	
	24				
	25	Luxury	Midsize	Sedan	
	24				
	...	...	...	...	..
	.				
	11481	Crossover	Compact	2dr SUV	
	26				
	11603	Crossover,Luxury,Performance	Midsize	4dr SUV	
	29				
	11604	Crossover,Luxury	Midsize	4dr SUV	
	30				
	11708	Crossover	Midsize	4dr SUV	
	22				
	11717	Crossover	Midsize	4dr SUV	
	22				
	city mpg	Popularity	MSRP		

14	19	3916	31500
18	17	3105	2000
20	17	3105	2000
24	17	3105	2000
25	17	3105	2000
...	...	...	...
11481	22	481	2000
11603	20	870	46350
11604	23	870	40950
11708	15	481	29149
11717	16	481	27499

[715 rows x 16 columns]

```
df.drop_duplicates(keep='first',inplace=True,ignore_index=True)
```

df

	Make	Model	Year	Fuel Type
Horse Power \				
0	BMW	1 Series M	2011	premium unleaded (required)
335.0				
1	BMW	1 Series	2011	premium unleaded (required)
300.0				
2	BMW	1 Series	2011	premium unleaded (required)
300.0				
3	BMW	1 Series	2011	premium unleaded (required)
230.0				
4	BMW	1 Series	2011	premium unleaded (required)
230.0				
...	...	...	...	...
...				
11194	Acura	ZDX	2012	premium unleaded (required)
300.0				
11195	Acura	ZDX	2012	premium unleaded (required)
300.0				
11196	Acura	ZDX	2012	premium unleaded (required)
300.0				
11197	Acura	ZDX	2013	premium unleaded (recommended)
300.0				
11198	Lincoln	Zephyr	2006	regular unleaded
221.0				

	Cylinders	Transmission	Type	Driven_Wheels	Number of Doors
\					
0	6.0		MANUAL	rear wheel drive	2.0
1	6.0		MANUAL	rear wheel drive	2.0
2	6.0		MANUAL	rear wheel drive	2.0

3	6.0	MANUAL	rear wheel drive	2.0
4	6.0	MANUAL	rear wheel drive	2.0
...	...	...	...	...
11194	6.0	AUTOMATIC	all wheel drive	4.0
11195	6.0	AUTOMATIC	all wheel drive	4.0
11196	6.0	AUTOMATIC	all wheel drive	4.0
11197	6.0	AUTOMATIC	all wheel drive	4.0
11198	6.0	AUTOMATIC	front wheel drive	4.0

Style \	Market Category	size	Vehicle
0	Factory Tuner,Luxury,High-Performance	Compact	Coupe
1	Luxury,Performance	Compact	Convertible
2	Luxury,High-Performance	Compact	Coupe
3	Luxury,Performance	Compact	Coupe
4	Luxury	Compact	Convertible
...	...	...	...
11194	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11195	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11196	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11197	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11198	Luxury	Midsize	Sedan

	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500
...	...	...	...	...
11194	23	16	204	46120

11195	23	16	204	56670
11196	23	16	204	50620
11197	23	16	204	50920
11198	26	17	61	28995

[11199 rows x 16 columns]

outliers

df.head()

	Make	Model	Year	Fuel Type	Horse Power
0	BMW	1 Series M	2011	premium unleaded (required)	335.0
1	BMW	1 Series	2011	premium unleaded (required)	300.0
2	BMW	1 Series	2011	premium unleaded (required)	300.0
3	BMW	1 Series	2011	premium unleaded (required)	230.0
4	BMW	1 Series	2011	premium unleaded (required)	230.0

	Transmission Type	Driven Wheels	Number of Doors
0	MANUAL	rear wheel drive	2.0
1	MANUAL	rear wheel drive	2.0
2	MANUAL	rear wheel drive	2.0
3	MANUAL	rear wheel drive	2.0
4	MANUAL	rear wheel drive	2.0

	Market Category	size	Vehicle Style
0	Factory Tuner,Luxury,High-Performance	Compact	Coupe
1	Luxury,Performance	Compact	Convertible
2	Luxury,High-Performance	Compact	Coupe
3	Luxury,Performance	Compact	Coupe
4	Luxury	Compact	Convertible

	city mpg	Popularity	MSRP
0	19	3916	46135
1	19	3916	40650
2	20	3916	36350
3	18	3916	29450
4	18	3916	34500

```
df[df['Horse Power']>1000]
```

	Make	Model	Year	Fuel Type	Horse Power
10661	Bugatti	Veyron 16.4	2008	premium unleaded (required)	1001.0
10662	Bugatti	Veyron 16.4	2008	premium unleaded (required)	1001.0
10663	Bugatti	Veyron 16.4	2009	premium unleaded (required)	1001.0

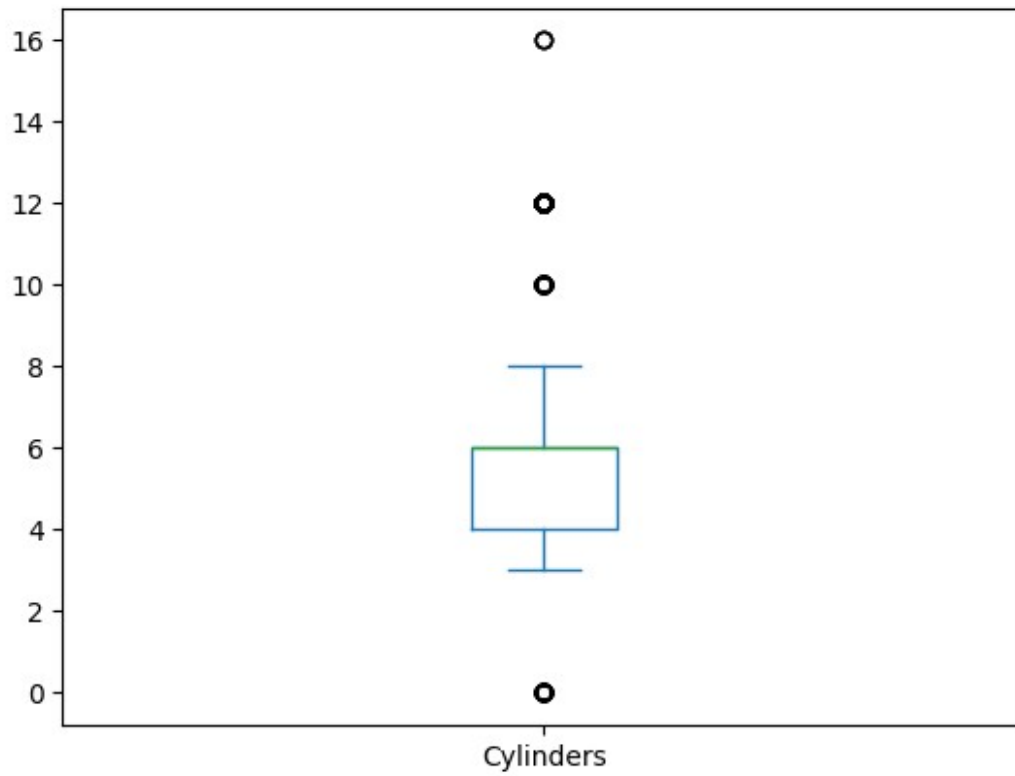
	Cylinders	Transmission	Type	Driven_Wheels	Number of Doors
10661	16.0	AUTOMATED_MANUAL		all wheel drive	2.0
10662	16.0	AUTOMATED_MANUAL		all wheel drive	2.0
10663	16.0	AUTOMATED_MANUAL		all wheel drive	2.0

	Market Category	size	Vehicle Style	highway MPG
10661	Exotic,High-Performance	Compact	Coupe	148
10662	Exotic,High-Performance	Compact	Coupe	148
10663	Exotic,High-Performance	Compact	Coupe	148

	Popularity	MSRP
10661	820	2065902
10662	820	1500000
10663	820	1705769

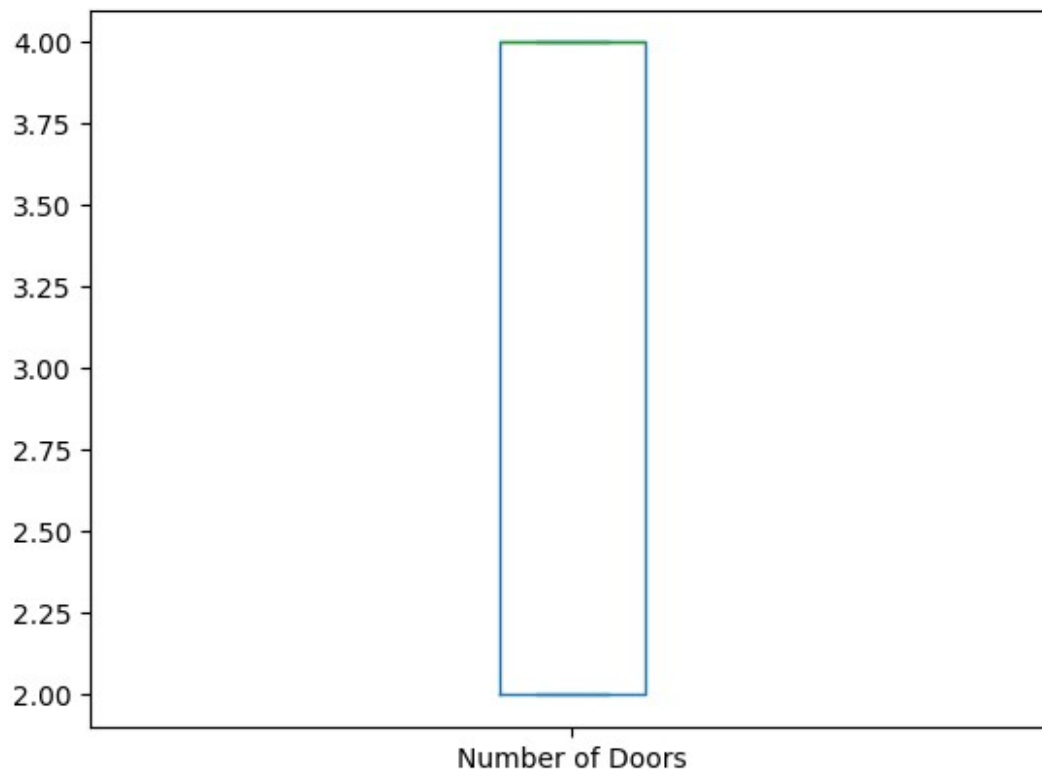
```
df['Cylinders'].plot(kind='box')
```

```
<Axes: >
```



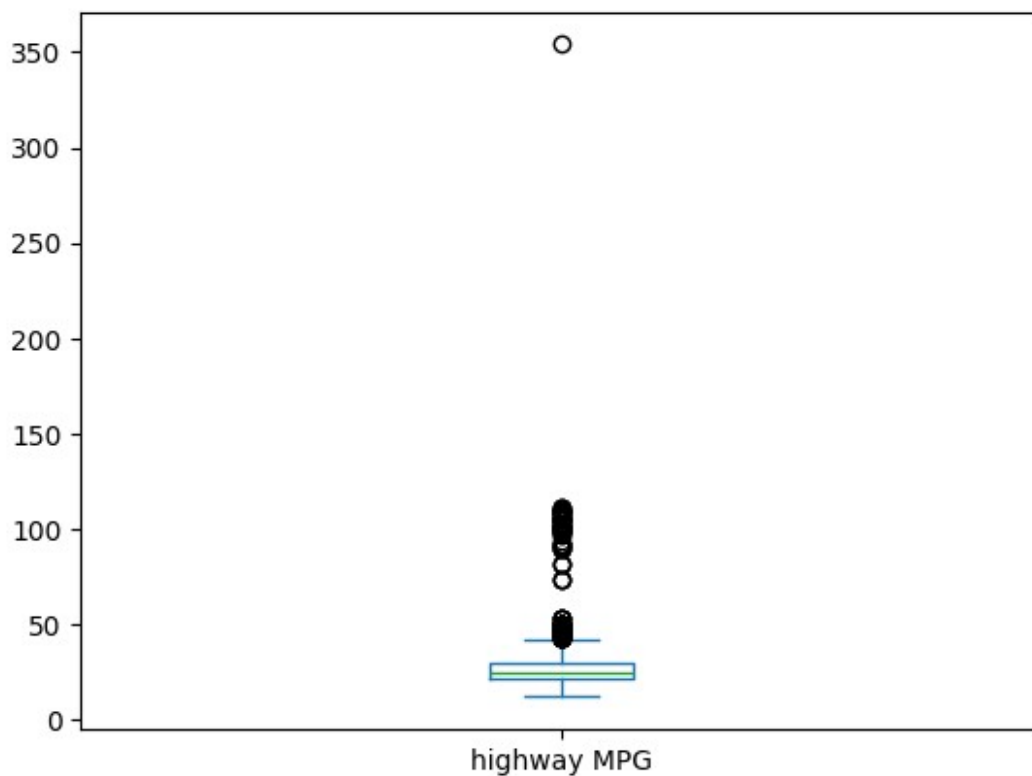
```
df['Number of Doors'].plot(kind='box')
```

```
<Axes: >
```



```
df['highway MPG'].plot(kind='box')
```

```
<Axes: >
```



```
df['highway MPG']>300
```

```
0      False
1      False
2      False
3      False
4      False
```

```
...
11194   False
11195   False
11196   False
11197   False
11198   False
```

```
Name: highway MPG, Length: 11199, dtype: bool
```

```
df[df['highway MPG']>300]
```

	Make	Model	Year	Fuel Type	Horse Power
1083	Audi	A6	2017	premium unleaded (recommended)	252.0

	Cylinders	Transmission Type	Driven_Wheels	Number of Doors
1083	4.0	AUTOMATED_MANUAL	front wheel drive	4.0

Market Category	size	Vehicle Style	highway MPG	city mpg
-----------------	------	---------------	-------------	----------

1083	Luxury	Midsize	Sedan	354	24
	Popularity	MSRP			
1083	3105	51600			
df					
	Make	Model	Year		Fuel Type
Horse Power \					
0	BMW	1 Series M	2011	premium unleaded (required)	
335.0					
1	BMW	1 Series	2011	premium unleaded (required)	
300.0					
2	BMW	1 Series	2011	premium unleaded (required)	
300.0					
3	BMW	1 Series	2011	premium unleaded (required)	
230.0					
4	BMW	1 Series	2011	premium unleaded (required)	
230.0					
...	...	...	...		...
...					
11194	Acura	ZDX	2012	premium unleaded (required)	
300.0					
11195	Acura	ZDX	2012	premium unleaded (required)	
300.0					
11196	Acura	ZDX	2012	premium unleaded (required)	
300.0					
11197	Acura	ZDX	2013	premium unleaded (recommended)	
300.0					
11198	Lincoln	Zephyr	2006	regular unleaded	
221.0					
	Cylinders	Transmission	Type	Driven_Wheels	Number of Doors
\					
0	6.0		MANUAL	rear wheel drive	2.0
1	6.0		MANUAL	rear wheel drive	2.0
2	6.0		MANUAL	rear wheel drive	2.0
3	6.0		MANUAL	rear wheel drive	2.0
4	6.0		MANUAL	rear wheel drive	2.0
...	...		...	...	...
11194	6.0		AUTOMATIC	all wheel drive	4.0
11195	6.0		AUTOMATIC	all wheel drive	4.0
11196	6.0		AUTOMATIC	all wheel drive	4.0

11197	6.0	AUTOMATIC	all wheel drive	4.0
11198	6.0	AUTOMATIC	front wheel drive	4.0

Style	Market Category	size	Vehicle
0	Factory Tuner,Luxury,High-Performance	Compact	Coupe
1	Luxury,Performance	Compact	Convertible
2	Luxury,High-Performance	Compact	Coupe
3	Luxury,Performance	Compact	Coupe
4	Luxury	Compact	Convertible
...	...	...	...
11194	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11195	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11196	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11197	Crossover,Hatchback,Luxury	Midsize	4dr Hatchback
11198	Luxury	Midsize	Sedan

	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500
...	...	...	...	...
11194	23	16	204	46120
11195	23	16	204	56670
11196	23	16	204	50620
11197	23	16	204	50920
11198	26	17	61	28995

[11199 rows x 16 columns]

```
df['highway MPG'].where(df['highway MPG']<300,30,inplace=True)
```

```
df
```

Horse Power \	Make	Model	Year	Fuel Type	
0	BMW	1 Series M	2011	premium unleaded (required)	
335.0					
1	BMW	1 Series	2011	premium unleaded (required)	
300.0					
2	BMW	1 Series	2011	premium unleaded (required)	
300.0					
3	BMW	1 Series	2011	premium unleaded (required)	
230.0					
4	BMW	1 Series	2011	premium unleaded (required)	
230.0					
...	...	...	...	...	
...					
11194	Acura	ZDX	2012	premium unleaded (required)	
300.0					
11195	Acura	ZDX	2012	premium unleaded (required)	
300.0					
11196	Acura	ZDX	2012	premium unleaded (required)	
300.0					
11197	Acura	ZDX	2013	premium unleaded (recommended)	
300.0					
11198	Lincoln	Zephyr	2006	regular unleaded	
221.0					
\	Cylinders	Transmission	Type	Driven_Wheels	Number of Doors
0	6.0		MANUAL	rear wheel drive	2.0
1	6.0		MANUAL	rear wheel drive	2.0
2	6.0		MANUAL	rear wheel drive	2.0
3	6.0		MANUAL	rear wheel drive	2.0
4	6.0		MANUAL	rear wheel drive	2.0
...	...		...	...	...
11194	6.0		AUTOMATIC	all wheel drive	4.0
11195	6.0		AUTOMATIC	all wheel drive	4.0
11196	6.0		AUTOMATIC	all wheel drive	4.0
11197	6.0		AUTOMATIC	all wheel drive	4.0
11198	6.0		AUTOMATIC	front wheel drive	4.0
			Market Category	size	Vehicle

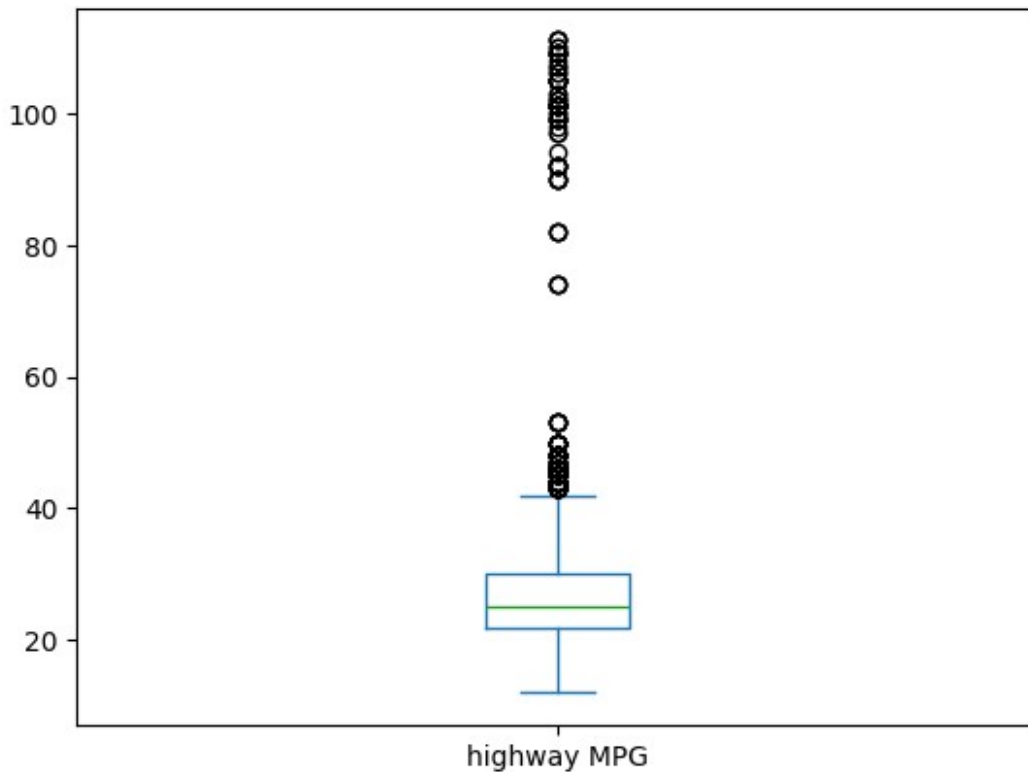
Style \				
0	Factory Tuner,Luxury,High-Performance	Compact		Coupe
1	Luxury,Performance	Compact		Convertible
2	Luxury,High-Performance	Compact		Coupe
3	Luxury,Performance	Compact		Coupe
4	Luxury	Compact		Convertible
...	...	...		...
11194	Crossover,Hatchback,Luxury	Midsize	4dr	Hatchback
11195	Crossover,Hatchback,Luxury	Midsize	4dr	Hatchback
11196	Crossover,Hatchback,Luxury	Midsize	4dr	Hatchback
11197	Crossover,Hatchback,Luxury	Midsize	4dr	Hatchback
11198	Luxury	Midsize		Sedan

	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500
...	...	...	...	...
11194	23	16	204	46120
11195	23	16	204	56670
11196	23	16	204	50620
11197	23	16	204	50920
11198	26	17	61	28995

[11199 rows x 16 columns]

```
df['highway MPG'].plot(kind='box')
```

<Axes: >



```
df['Cylinders']=df['Cylinders'].astype(np.uint8)
df
```

	Make	Model	Year	Fuel Type
Horse Power \				
0	BMW	1 Series M	2011	premium unleaded (required)
335.0				
1	BMW	1 Series	2011	premium unleaded (required)
300.0				
2	BMW	1 Series	2011	premium unleaded (required)
300.0				
3	BMW	1 Series	2011	premium unleaded (required)
230.0				
4	BMW	1 Series	2011	premium unleaded (required)
230.0				
...	...	...	...	...
...				
11194	Acura	ZDX	2012	premium unleaded (required)
300.0				
11195	Acura	ZDX	2012	premium unleaded (required)
300.0				
11196	Acura	ZDX	2012	premium unleaded (required)
300.0				
11197	Acura	ZDX	2013	premium unleaded (recommended)
300.0				

11198	Lincoln	Zephyr	2006	regular unleaded		
221.0						
\	Cylinders	Transmission	Type	Driven_Wheels	Number of Doors	
	0	6	MANUAL	rear wheel drive	2.0	
	1	6	MANUAL	rear wheel drive	2.0	
	2	6	MANUAL	rear wheel drive	2.0	
	3	6	MANUAL	rear wheel drive	2.0	
	4	6	MANUAL	rear wheel drive	2.0	
...	...	...		...	...	
11194	6	AUTOMATIC	all wheel drive	4.0		
11195	6	AUTOMATIC	all wheel drive	4.0		
11196	6	AUTOMATIC	all wheel drive	4.0		
11197	6	AUTOMATIC	all wheel drive	4.0		
11198	6	AUTOMATIC	front wheel drive	4.0		
Style \	Market Category			size	Vehicle	
	0	Factory Tuner,Luxury,High-Performance		Compact	Coupe	
	1	Luxury,Performance		Compact	Convertible	
	2	Luxury,High-Performance		Compact	Coupe	
	3	Luxury,Performance		Compact	Coupe	
	4	Luxury		Compact	Convertible	
...	...		...	...		
11194	Crossover,Hatchback,Luxury			Midsize	4dr Hatchback	
11195	Crossover,Hatchback,Luxury			Midsize	4dr Hatchback	
11196	Crossover,Hatchback,Luxury			Midsize	4dr Hatchback	
11197	Crossover,Hatchback,Luxury			Midsize	4dr Hatchback	
11198	Luxury			Midsize	Sedan	

	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500
...	...	...	...	...
11194	23	16	204	46120
11195	23	16	204	56670
11196	23	16	204	50620
11197	23	16	204	50920
11198	26	17	61	28995

[11199 rows x 16 columns]

df

	Make	Model	Year	Fuel Type
Horse Power \				
0	BMW	1 Series M	2011	premium unleaded (required)
335.0				
1	BMW	1 Series	2011	premium unleaded (required)
300.0				
2	BMW	1 Series	2011	premium unleaded (required)
300.0				
3	BMW	1 Series	2011	premium unleaded (required)
230.0				
4	BMW	1 Series	2011	premium unleaded (required)
230.0				
...	...	...	...	...
...				
11194	Acura	ZDX	2012	premium unleaded (required)
300.0				
11195	Acura	ZDX	2012	premium unleaded (required)
300.0				
11196	Acura	ZDX	2012	premium unleaded (required)
300.0				
11197	Acura	ZDX	2013	premium unleaded (recommended)
300.0				
11198	Lincoln	Zephyr	2006	regular unleaded
221.0				

	Cylinders	Transmission Type	Driven_Wheels	Number of Doors
\				
0	6	MANUAL	rear wheel drive	2.0
1	6	MANUAL	rear wheel drive	2.0

2	6	MANUAL	rear wheel drive	2.0
3	6	MANUAL	rear wheel drive	2.0
4	6	MANUAL	rear wheel drive	2.0
...	...	...	...	...
11194	6	AUTOMATIC	all wheel drive	4.0
11195	6	AUTOMATIC	all wheel drive	4.0
11196	6	AUTOMATIC	all wheel drive	4.0
11197	6	AUTOMATIC	all wheel drive	4.0
11198	6	AUTOMATIC	front wheel drive	4.0
		Market Category	size	Vehicle
Style \				
0	Factory Tuner,Luxury,High-Performance		Compact	Coupe
1	Luxury,Performance		Compact	Convertible
2	Luxury,High-Performance		Compact	Coupe
3	Luxury,Performance		Compact	Coupe
4	Luxury		Compact	Convertible
...	...		...	...
11194	Crossover,Hatchback,Luxury		Midsize	4dr Hatchback
11195	Crossover,Hatchback,Luxury		Midsize	4dr Hatchback
11196	Crossover,Hatchback,Luxury		Midsize	4dr Hatchback
11197	Crossover,Hatchback,Luxury		Midsize	4dr Hatchback
11198	Luxury		Midsize	Sedan
	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500
...	...	...	...	...

11194	23	16	204	46120
11195	23	16	204	56670
11196	23	16	204	50620
11197	23	16	204	50920
11198	26	17	61	28995

[11199 rows x 16 columns]

df.columns

```
Index(['Make', 'Model', 'Year', 'Fuel Type', 'Horse Power',
      'Cylinders',
      'Transmission Type', 'Driven_Wheels', 'Number of Doors',
      'Market Category', 'size', 'Vehicle Style', 'highway MPG',
      'city mpg',
      'Popularity', 'MSRP'],
      dtype='object')
```

df['Vehicle Style'].unique()

```
array(['Coupe', 'Convertible', 'Sedan', 'Wagon', '4dr Hatchback',
      '2dr Hatchback', '4dr SUV', 'Passenger Minivan', 'Cargo
      Minivan',
      'Crew Cab Pickup', 'Regular Cab Pickup', 'Extended Cab Pickup',
      '2dr SUV', 'Cargo Van', 'Convertible SUV', 'Passenger Van'],
      dtype=object)
```

```
def fun(n):
    return n.split()[-1]
```

df['Vehicle ']=df['Vehicle Style'].apply(fun)

df['Vehicle Style'].unique()

```
array(['Coupe', 'Convertible', 'Sedan', 'Wagon', '4dr Hatchback',
      '2dr Hatchback', '4dr SUV', 'Passenger Minivan', 'Cargo
      Minivan',
      'Crew Cab Pickup', 'Regular Cab Pickup', 'Extended Cab Pickup',
      '2dr SUV', 'Cargo Van', 'Convertible SUV', 'Passenger Van'],
      dtype=object)
```

df.isna().sum()

Make	0
Model	0
Year	0
Fuel Type	0
Horse Power	69
Cylinders	0
Transmission Type	0
Driven_Wheels	0
Number of Doors	0

```

Market Category      0
size                  0
Vehicle Style         0
highway MPG           0
city mpg              0
Popularity            0
MSRP                  0
Vehicle              0
dtype: int64

```

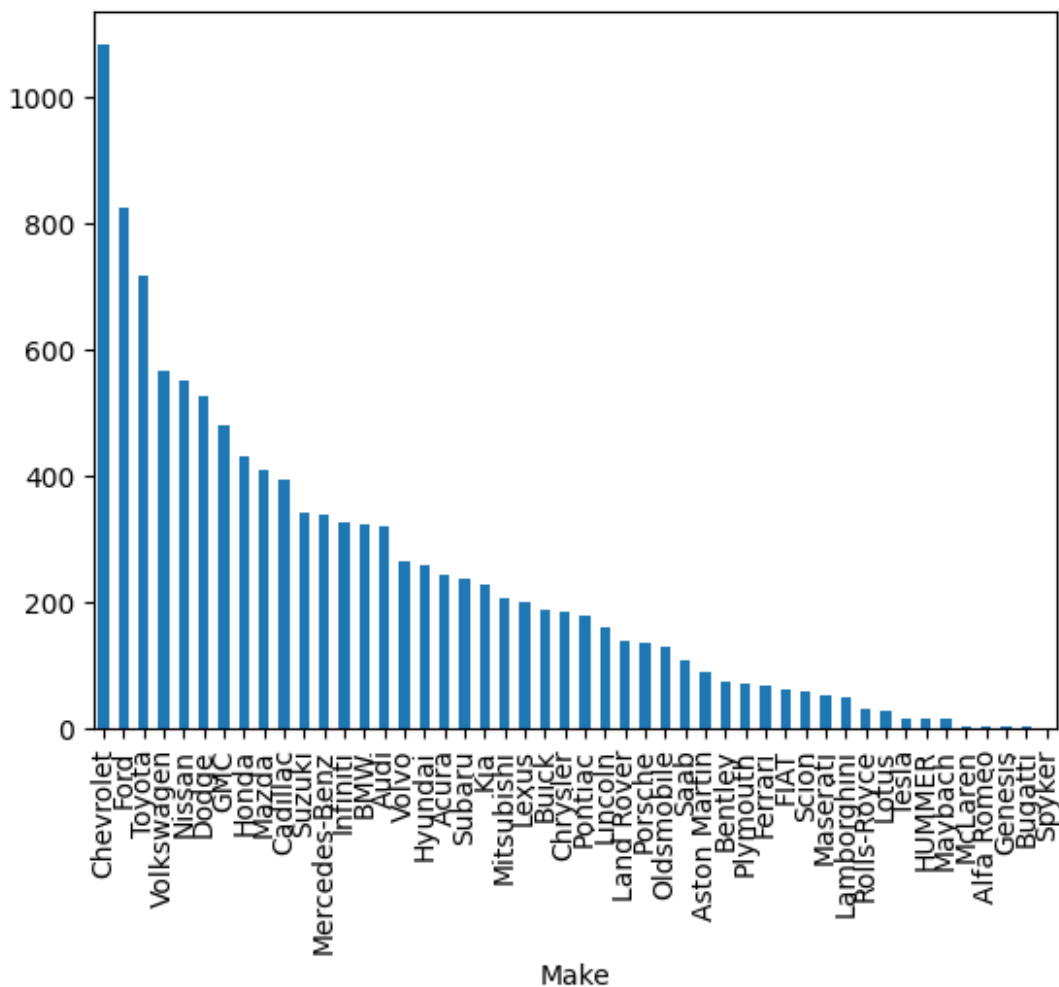
UNIVARIATE ANALYSIS

```

df.shape
(11199, 17)

df['Make'].value_counts().plot(kind='bar')
<Axes: xlabel='Make'>

```

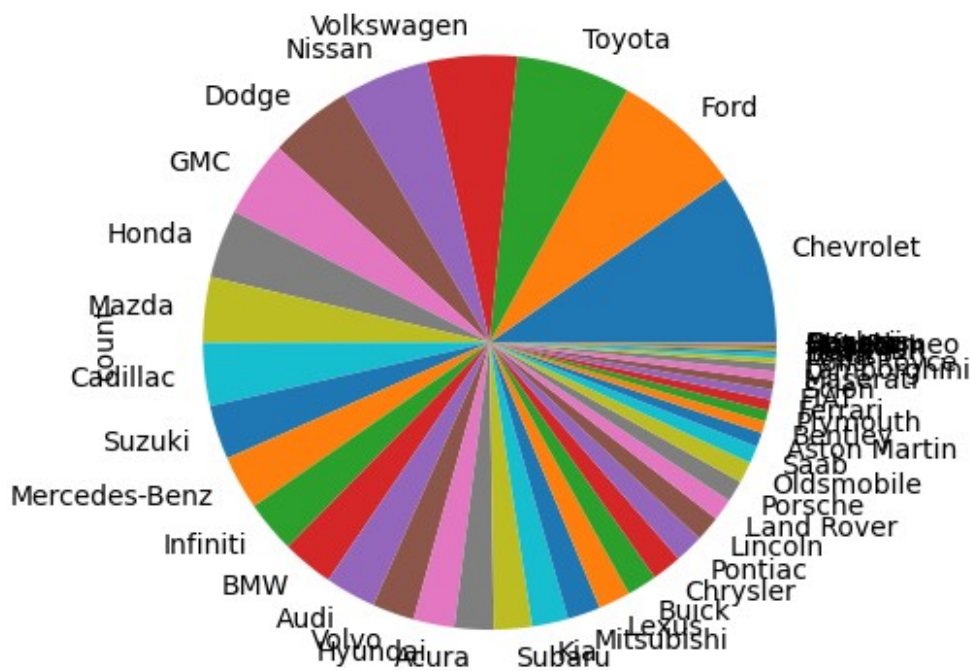


```
##INSIGHTS
```

```
## CHEVERLOET , FORD, AND TOYOTA HAS MORE NO OF CARS IN GIVEN DATASET  
## SPYKER, BUGATTI, MCLAREN , ALFA ROMEO HAS LESS NO OF CARS IN GIVEN  
DATA SET
```

```
df['Make'].value_counts().plot(kind='pie')
```

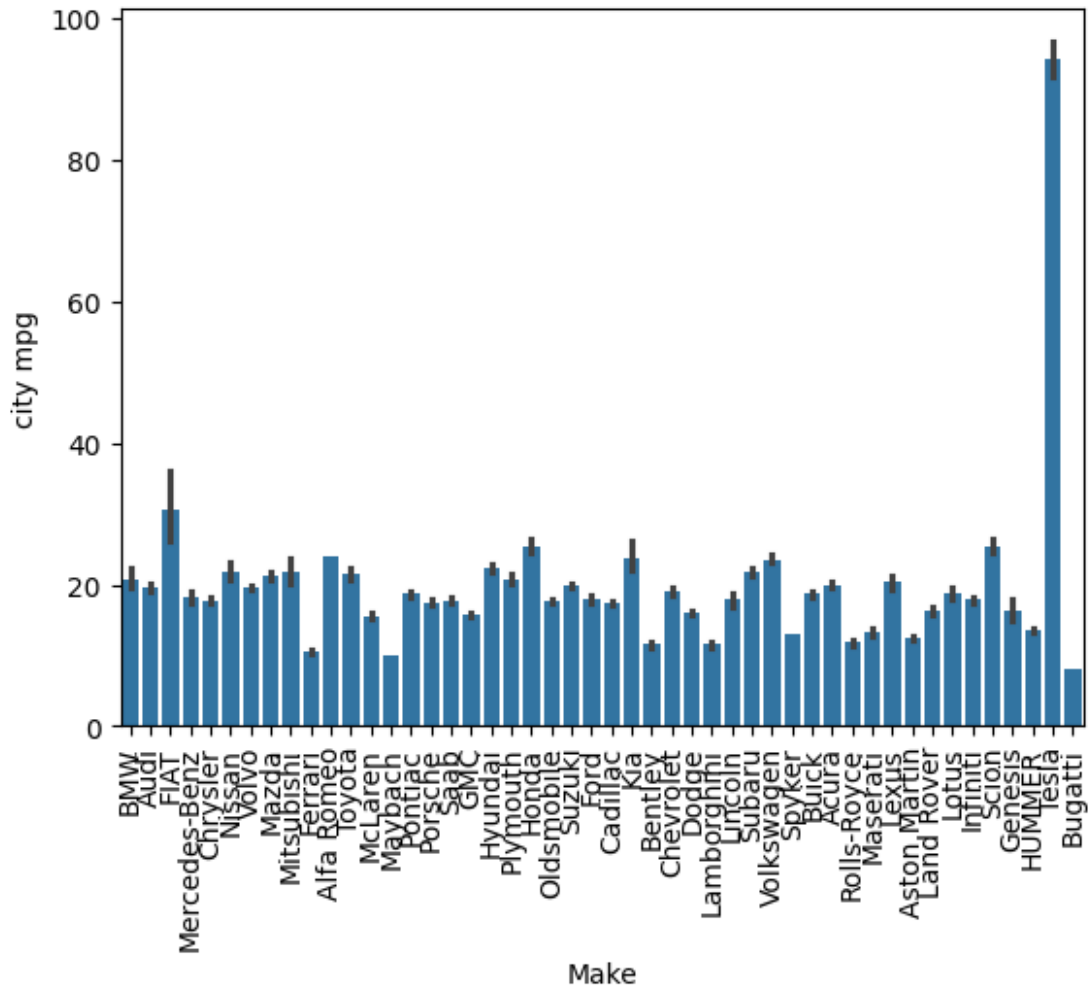
```
<Axes: ylabel='count'>
```



```
import seaborn as sns  
import matplotlib.pyplot as plt
```

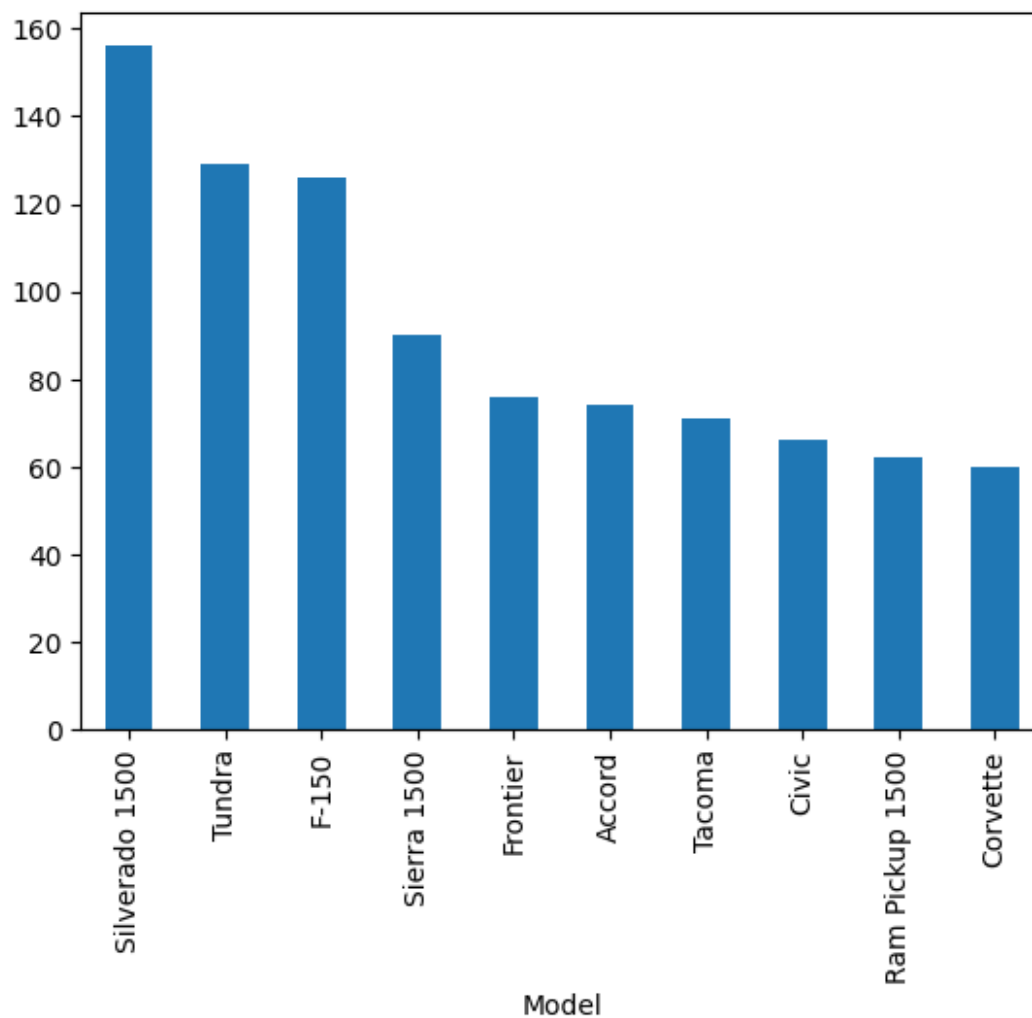
```
sns.barplot(data=df, x='Make', y='city mpg')
```

```
plt.xticks(rotation=90)  
plt.show()
```

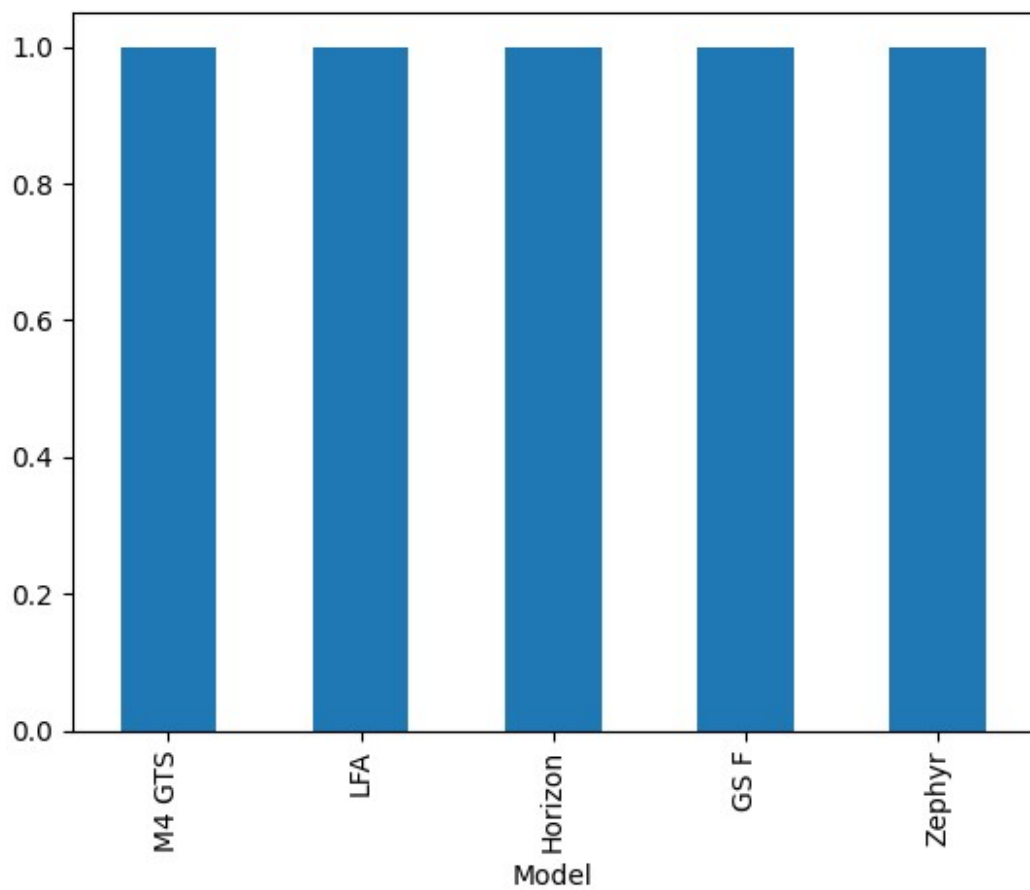



```
df['Model'].value_counts().head(10).plot(kind='bar')    ### to find top
10 data we use value counts
```

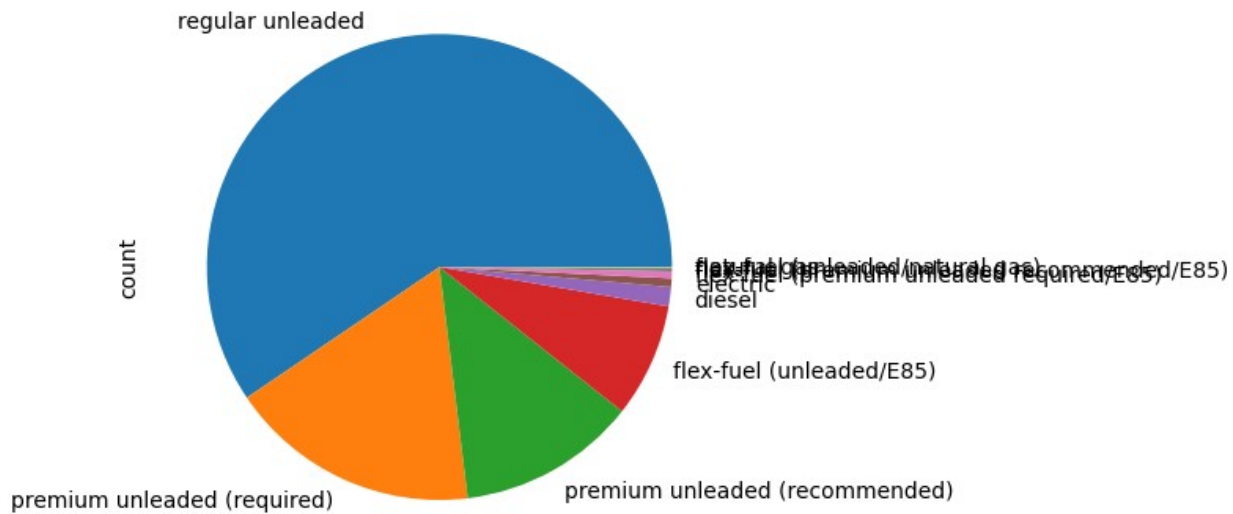
```
<Axes: xlabel='Model'>
```



```
df['Model'].value_counts().tail(5).plot(kind='bar')  
<Axes: xlabel='Model'>
```

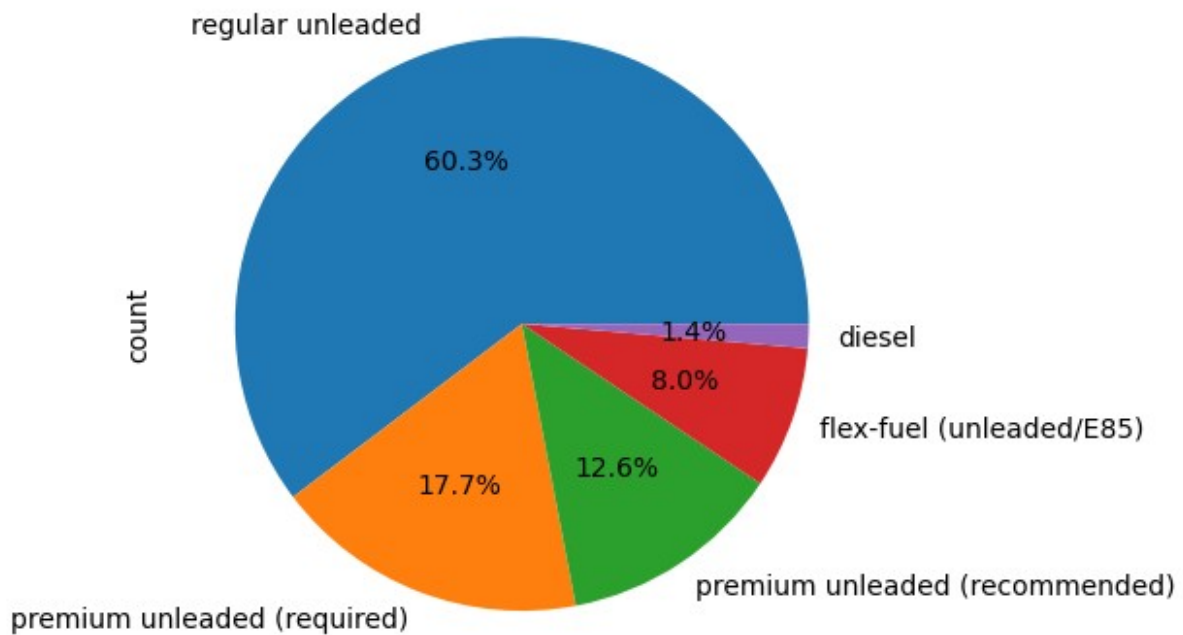


```
df['Fuel Type'].value_counts().plot(kind='pie')
<Axes: ylabel='count'>
```



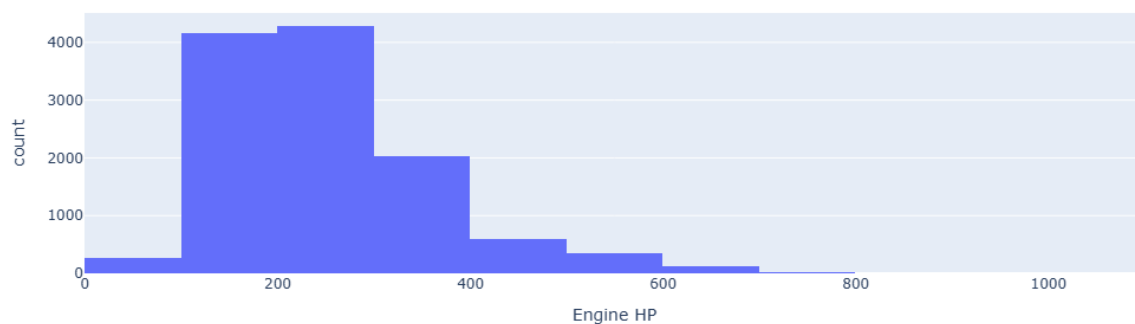
```
df['Fuel Type'].value_counts().head(5).plot(kind='pie',autopct = "%.1f%")
```

```
<Axes: ylabel='count'>
```



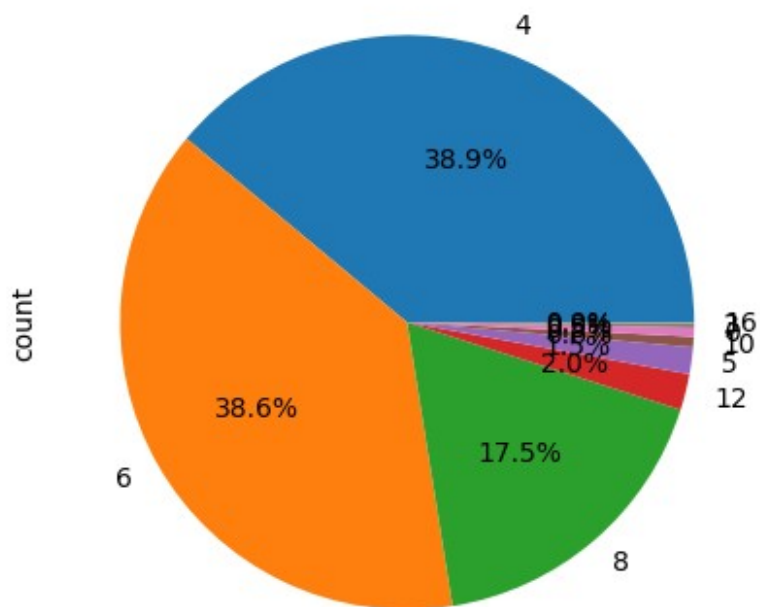
```
import plotly.express as px
```

```
px.histogram(data_frame=df,x='Engine HP',nbins=10)
```

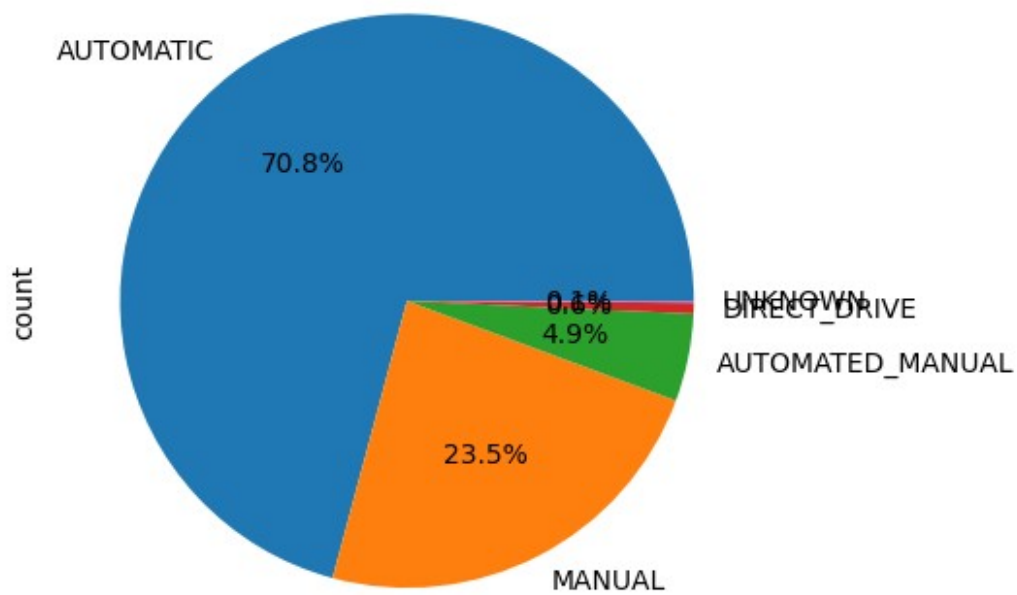


```
df['Cylinders'].value_counts().plot(kind='pie',autopct = "%.1f%%")
```

```
<Axes: ylabel='count'>
```



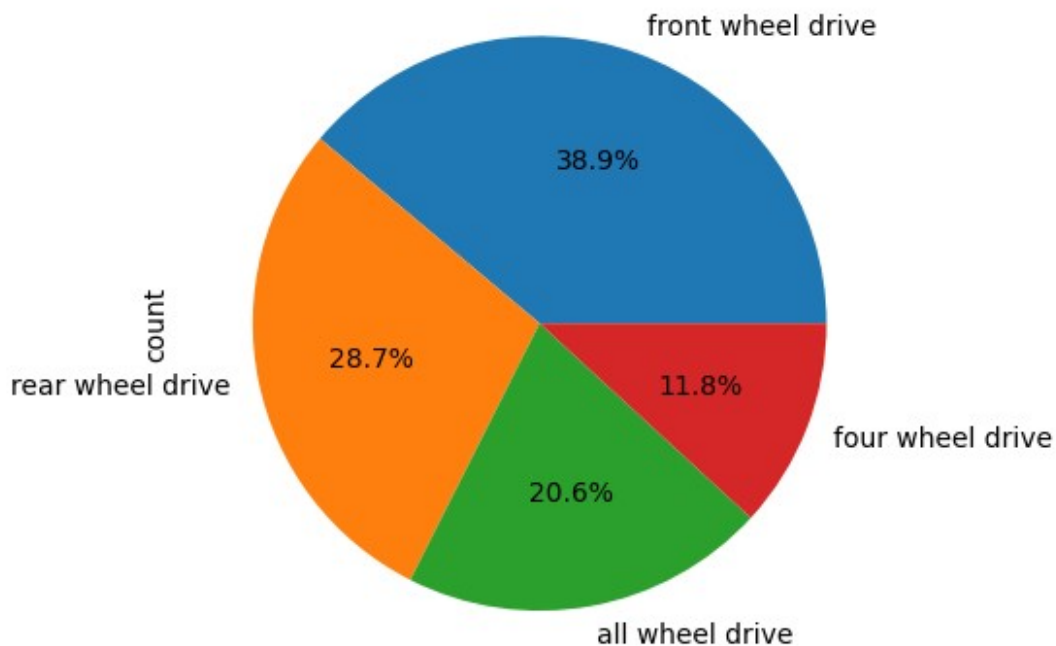
```
df['Transmission Type'].value_counts().plot(kind='pie', autopct = "%.1f%")
<Axes: ylabel='count'>
```



```
## 70% of cars belongs to automatic while 23% of cars belongs to manual.

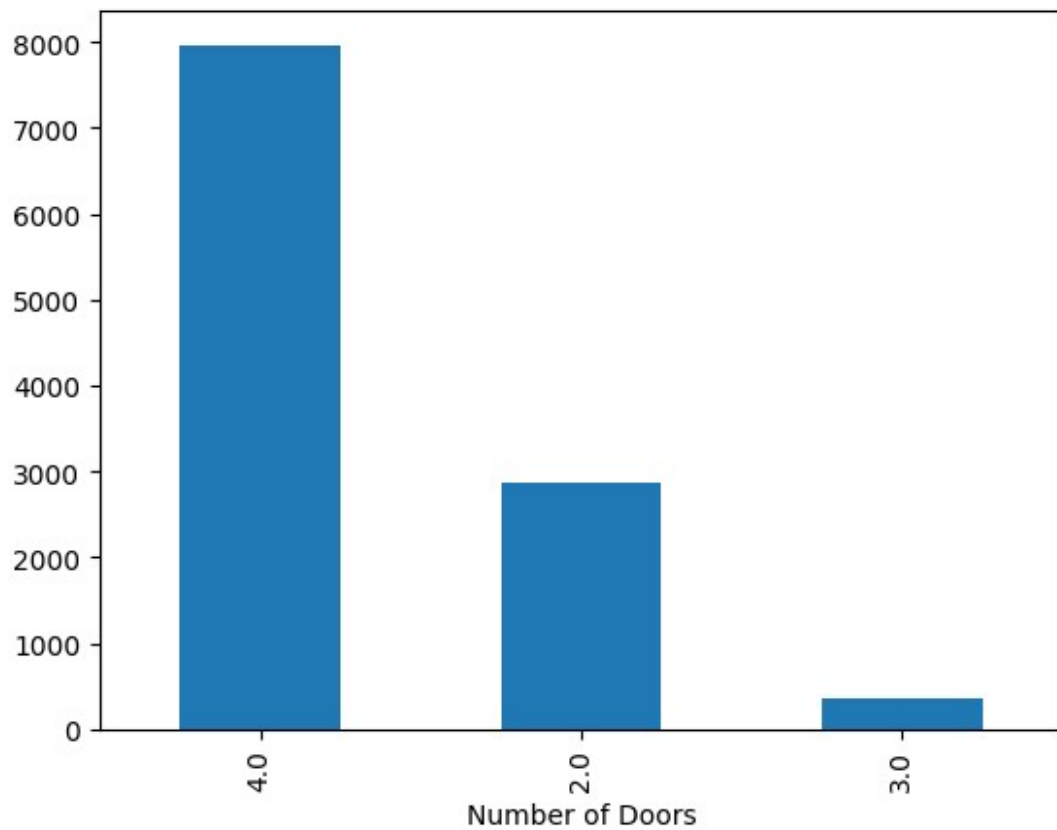
df.columns
Index(['Make', 'Model', 'Year', 'Fuel Type', 'Horse Power',
      'Cylinders',
      'Transmission Type', 'Driven_Wheels', 'Number of Doors',
      'Market Category', 'size', 'Vehicle Style', 'highway MPG',
      'city mpg',
      'Popularity', 'MSRP', 'Vehicle '],
      dtype='object')

df['Driven_Wheels'].value_counts().plot(kind='pie', autopct = "%.1f%%")
<Axes: ylabel='count'>
```



```
## more than 39% cars having front whhel drive
## more than 28% cars are running with rear wheel drive
## and least 11% cars are running with four wheel drive

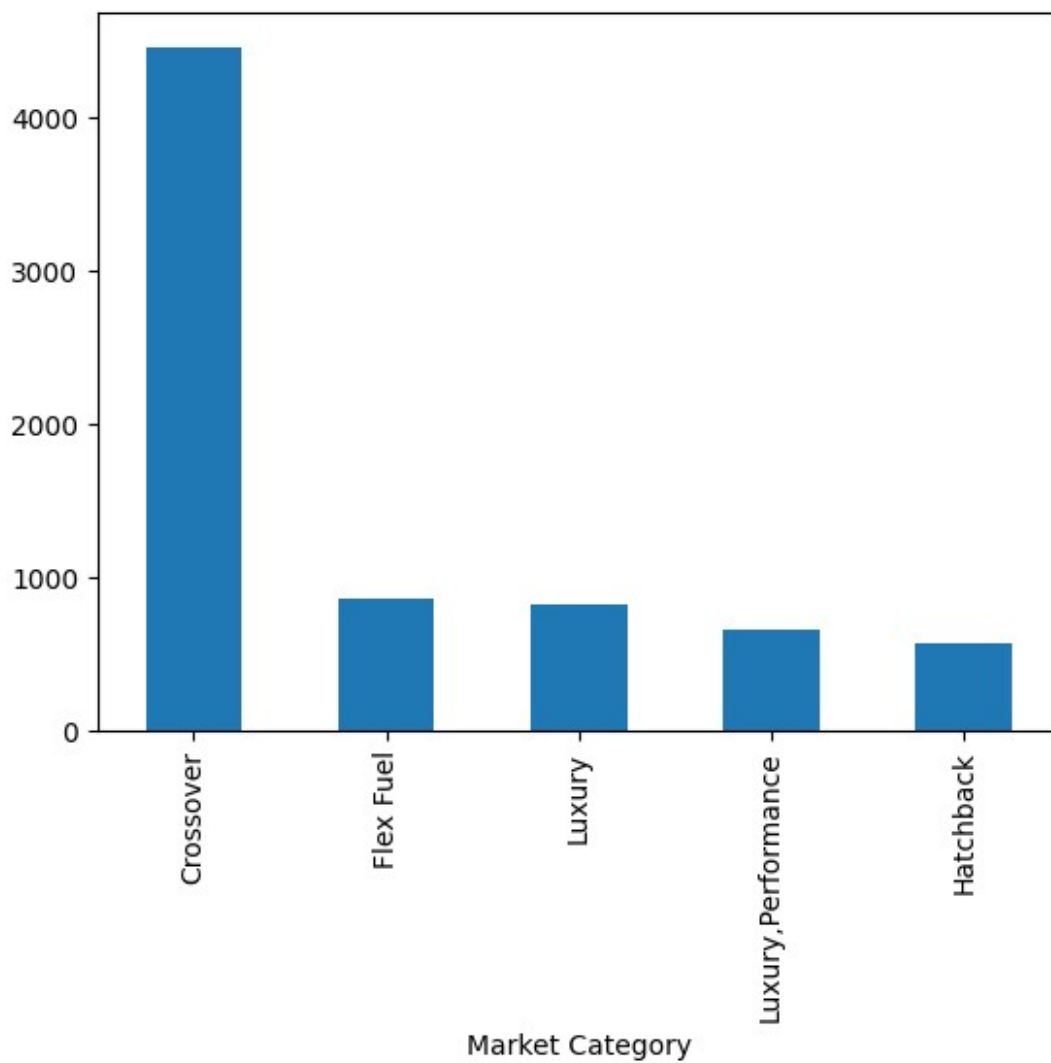
df['Number of Doors'].value_counts().plot(kind='bar')
<Axes: xlabel='Number of Doors'>
```



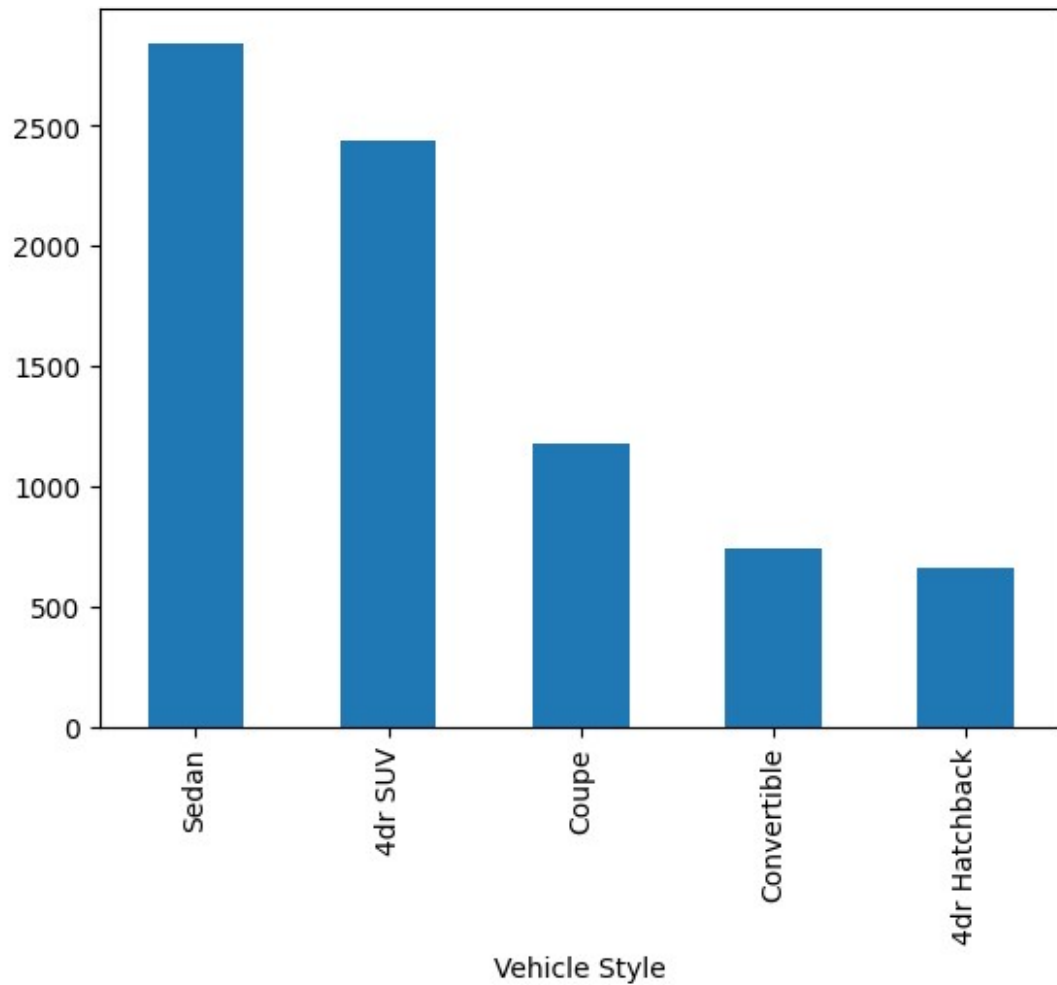
```
### around 8000 cars have 4 doors
```

```
df['Market Category'].value_counts().head(5).plot(kind='bar')
```

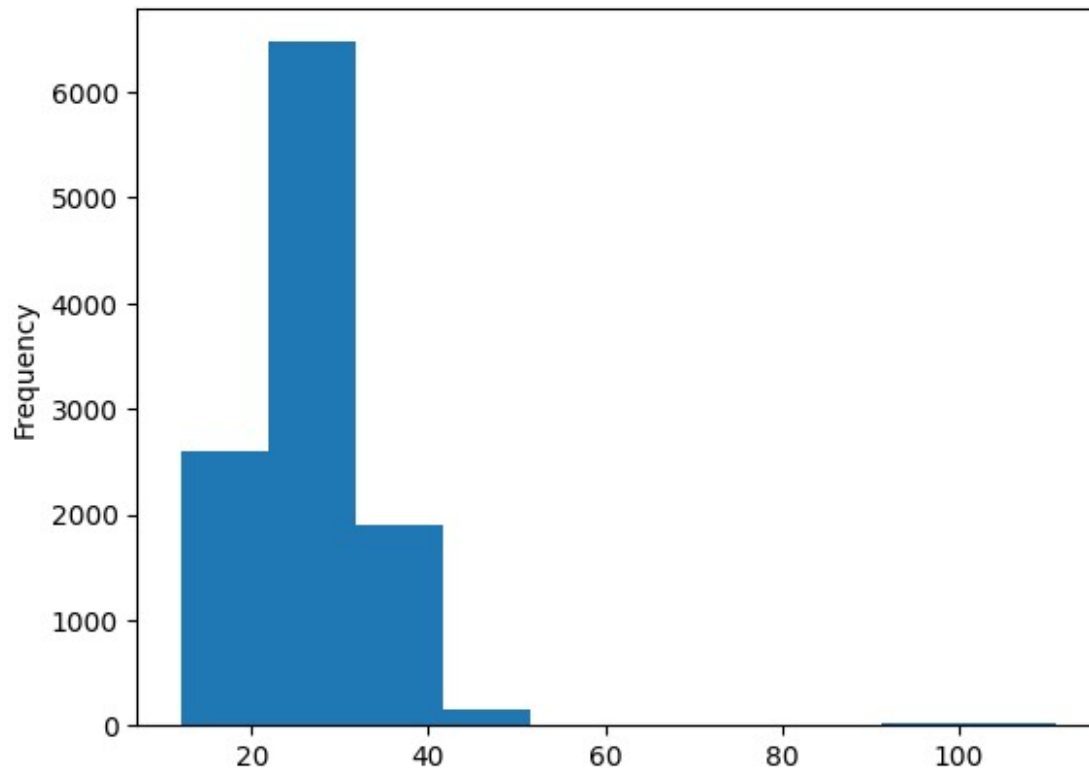
```
<Axes: xlabel='Market Category'>
```



```
###  
df['Vehicle Style'].value_counts().head(5).plot(kind='bar')  
<Axes: xlabel='Vehicle Style'>
```

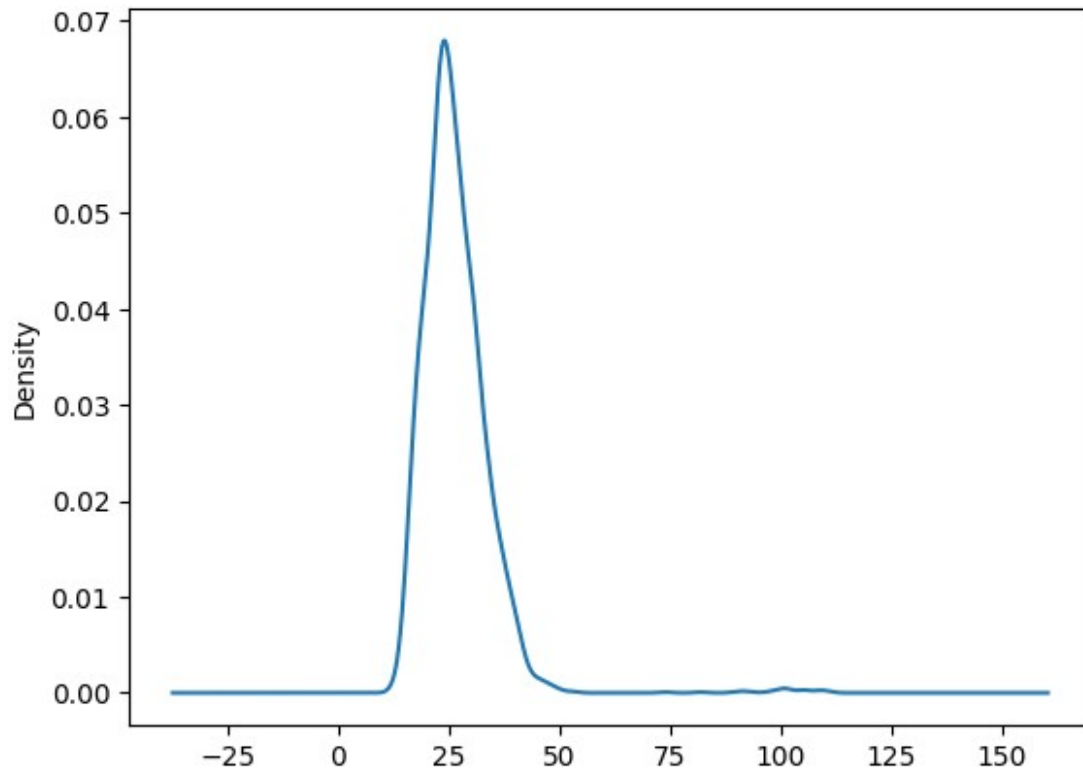
```
## there are more number of cars from sedan and SUV.  
df['highway MPG'].plot(kind='hist')  
<Axes: ylabel='Frequency'>
```



most of the cars mileage ranges from 10 to 35 in cities.

```
df['highway MPG'].plot(kind='kde')
```

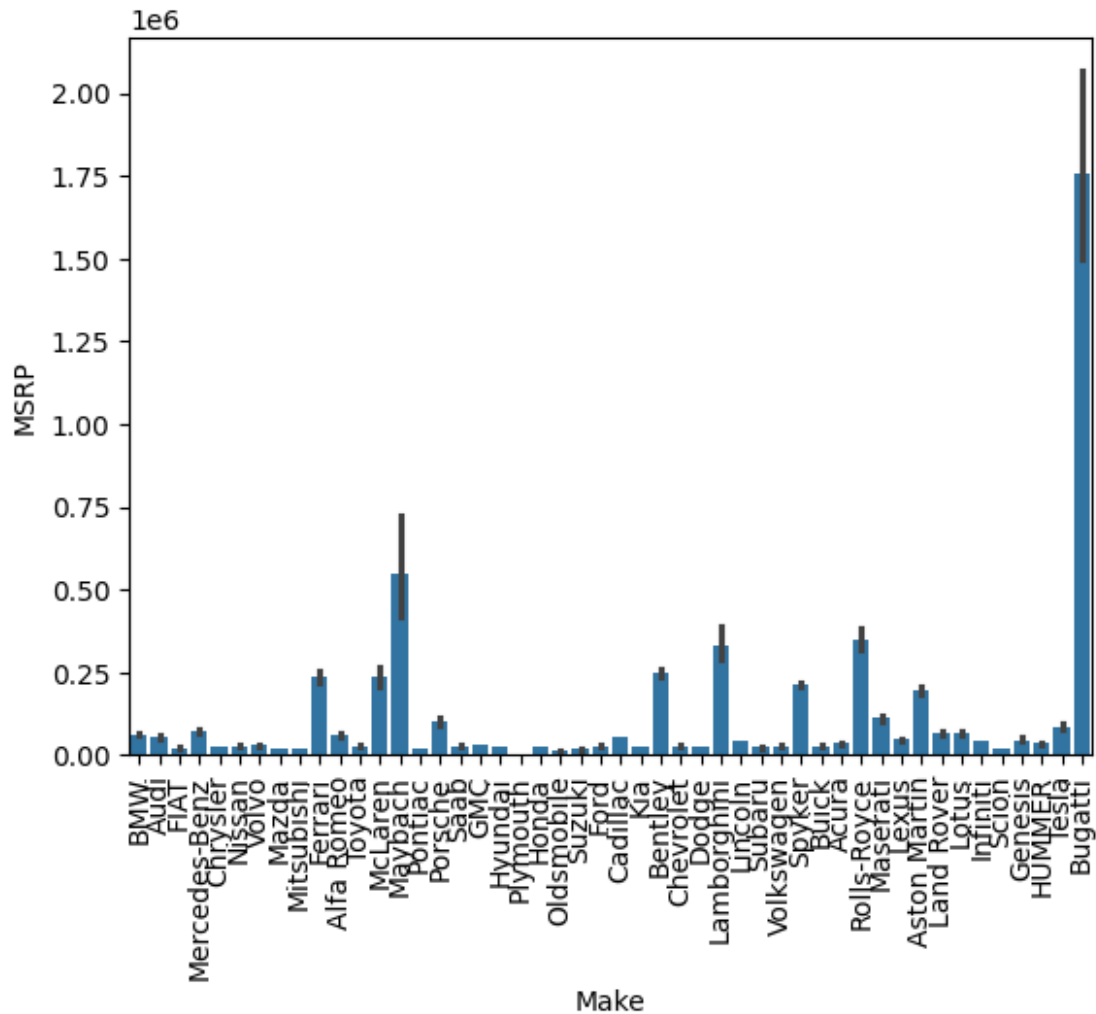
```
<Axes: ylabel='Density'>
```



```
## most of the cars mileage ranges from 20 to 40 in highways.  
df['MSRP'].mean() ##average price of a car in the given data set  
41925.92713635146  
  
df['MSRP'].max()    ## max price of car in the data set  
2065902  
  
df['MSRP'].min()## min price of a car in the given data set  
2000
```

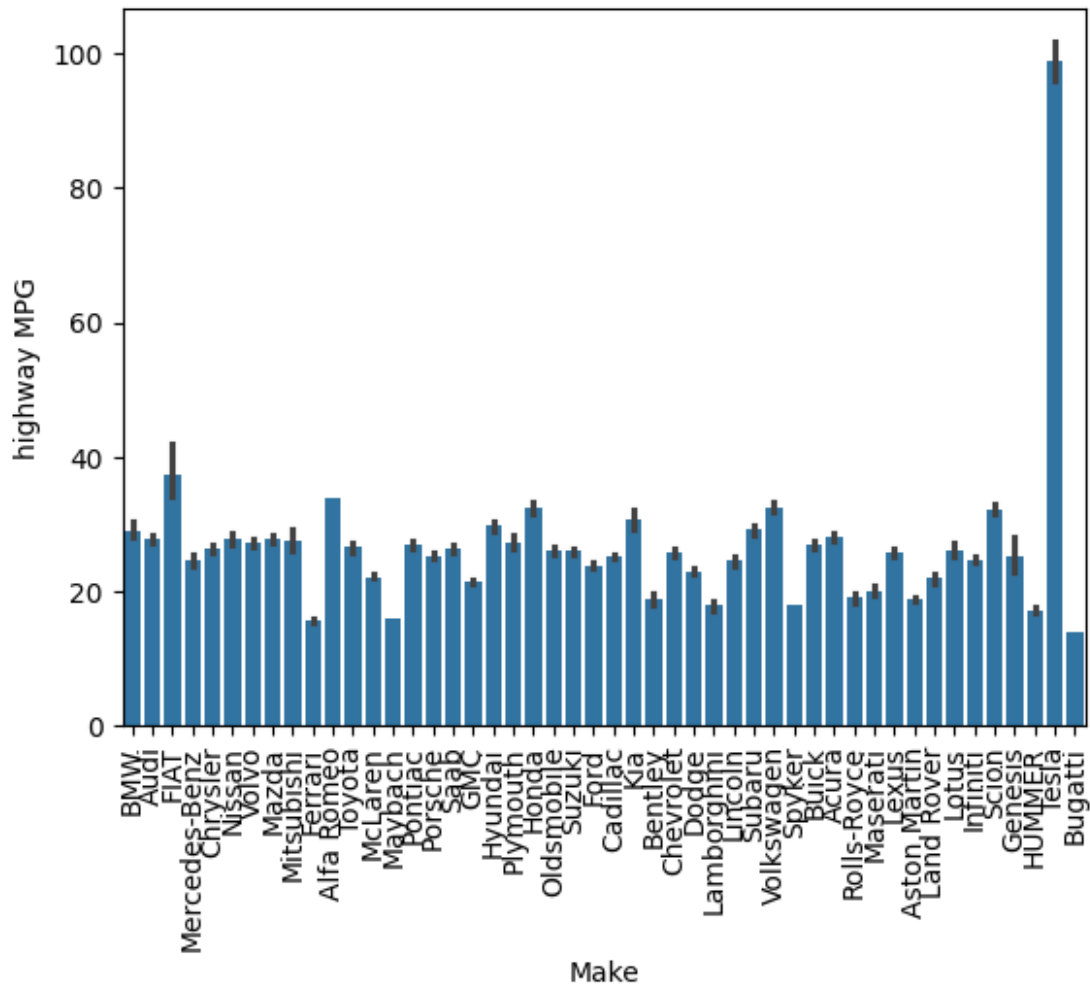
BIVARIATE ANALYSIS

```
sns.barplot(data=df,x='Make',y='MSRP')  
plt.xticks(rotation=90)  
plt.show()
```



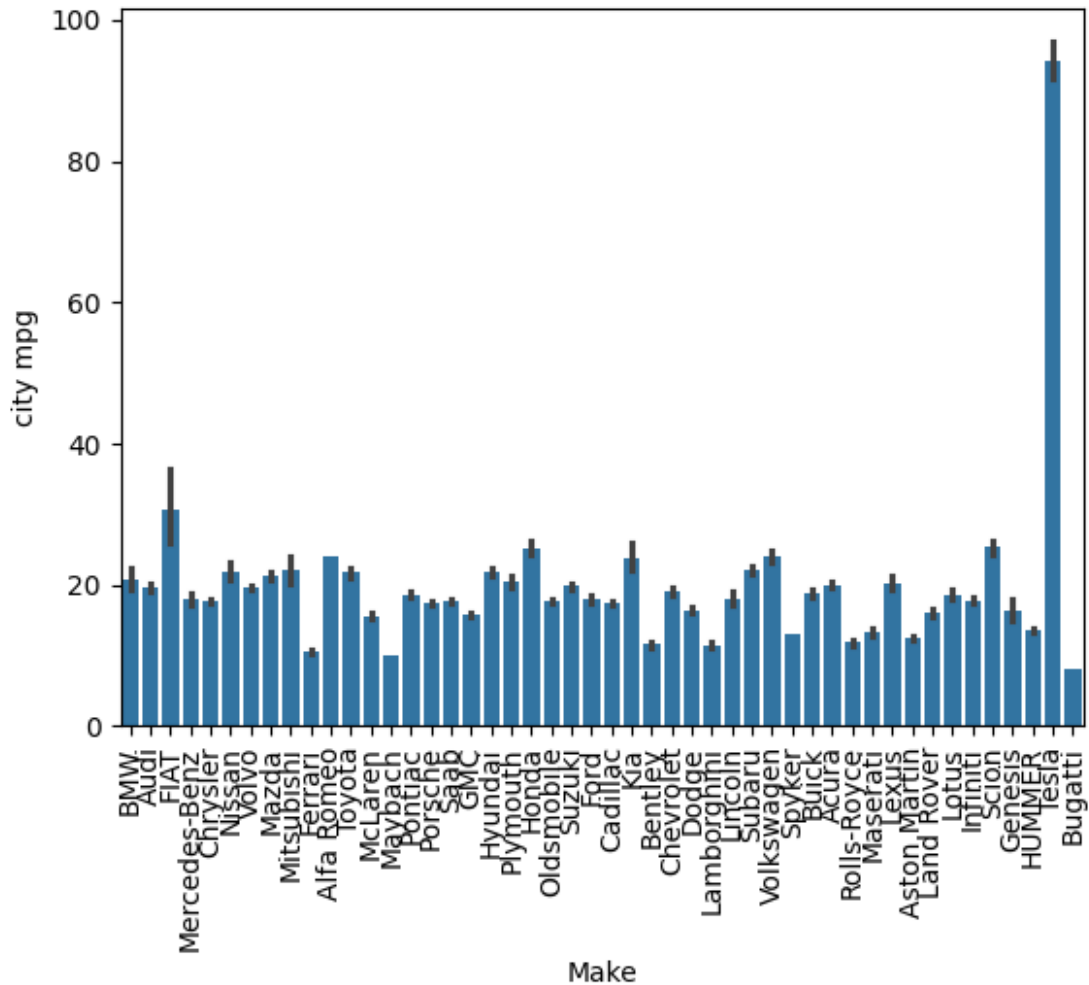
most expensive cars in the given dataset is
Bugatti, Maybach, lamborghini and rolls-royce

```
sns.barplot(data=df, x='Make', y='highway MPG')
plt.xticks(rotation=90)
plt.show()
```

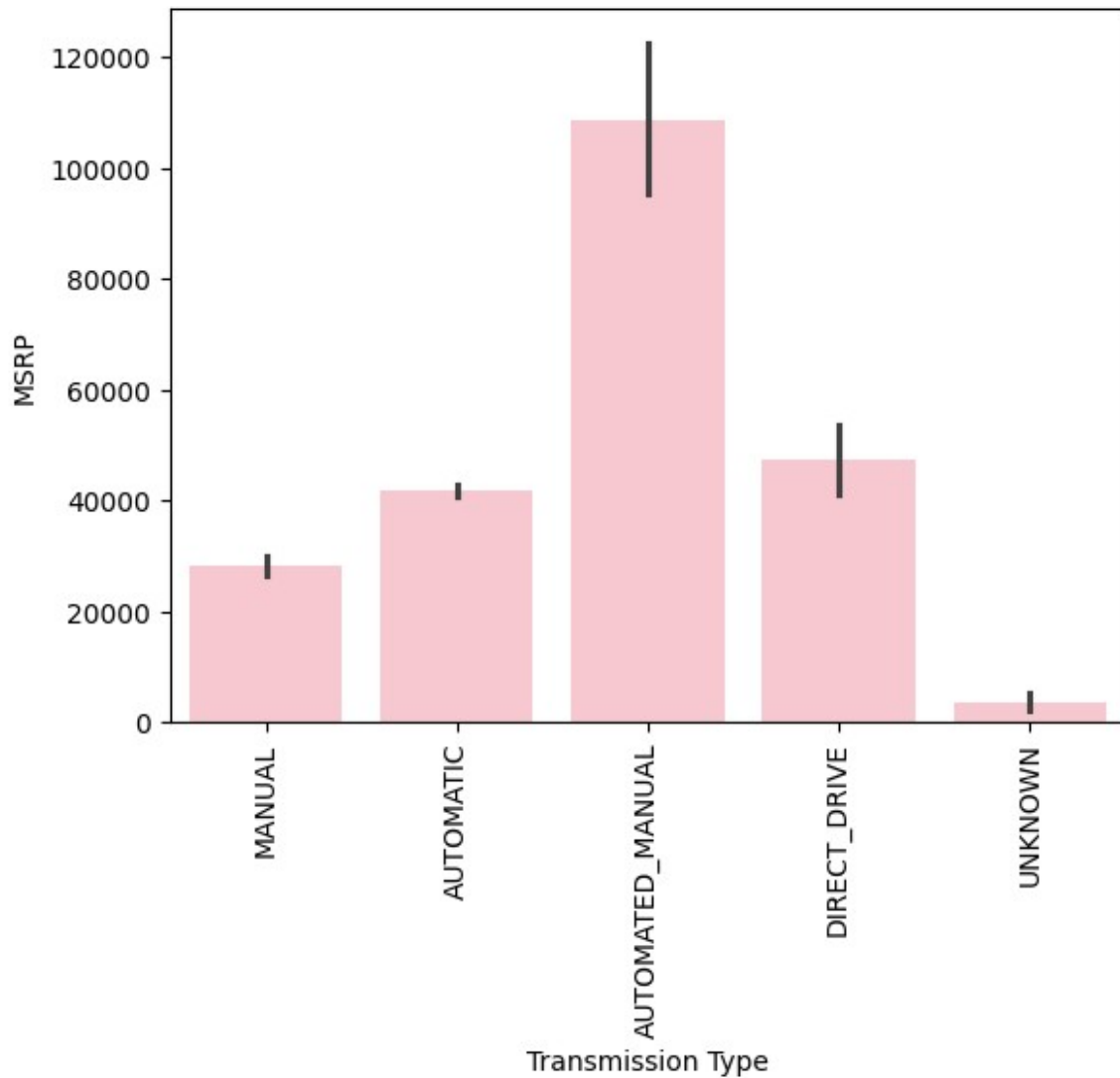


based on avg mileage best car brands are -
 tesla,fiat,volkswagen,kia,alfa romeo
 ##worst brands bugatti,ferari,maybach,hummer,spyker.

```
sns.barplot(data=df,x='Make',y='city mpg')
plt.xticks(rotation=90)
plt.show()
```

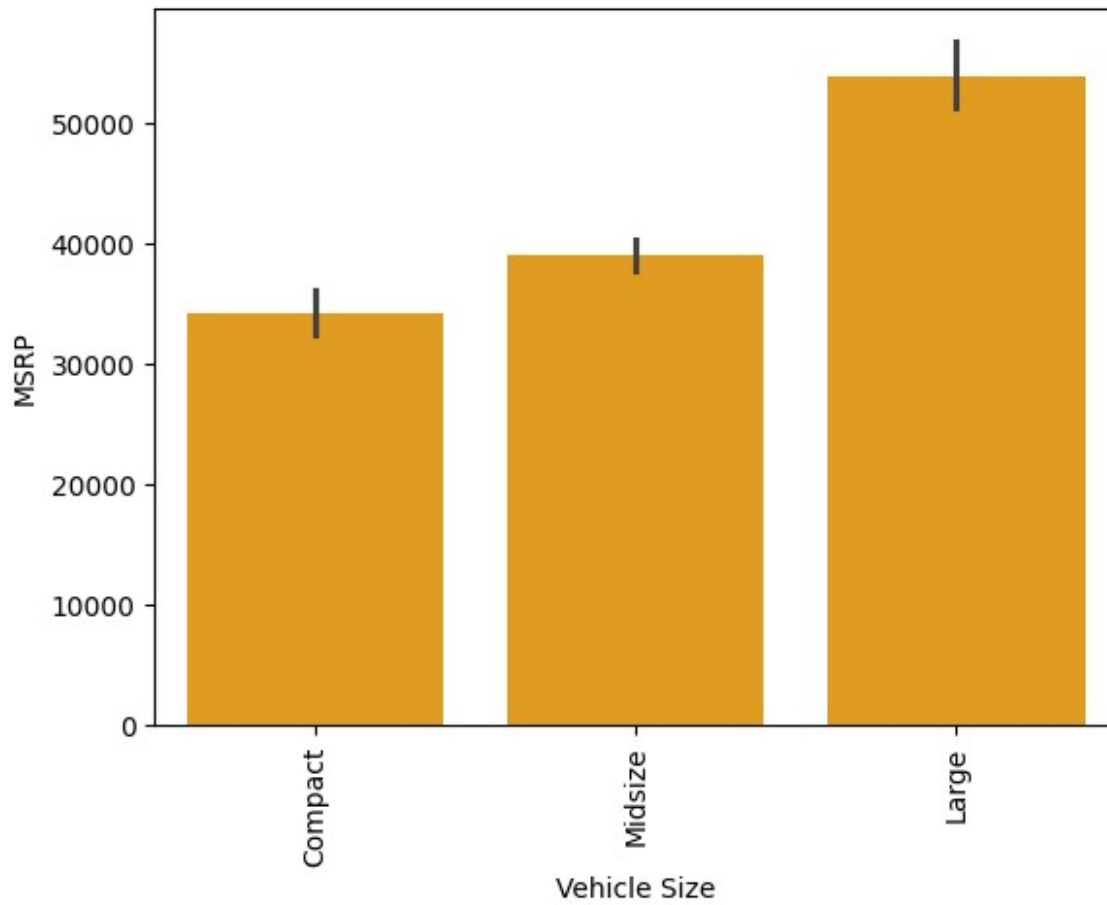


```
## noc hange in the mileage with respect to highway and city
sns.barplot(data=df,x='Transmission Type',y='MSRP' ,color='pink')
plt.xticks(rotation=90)
plt.show()
```

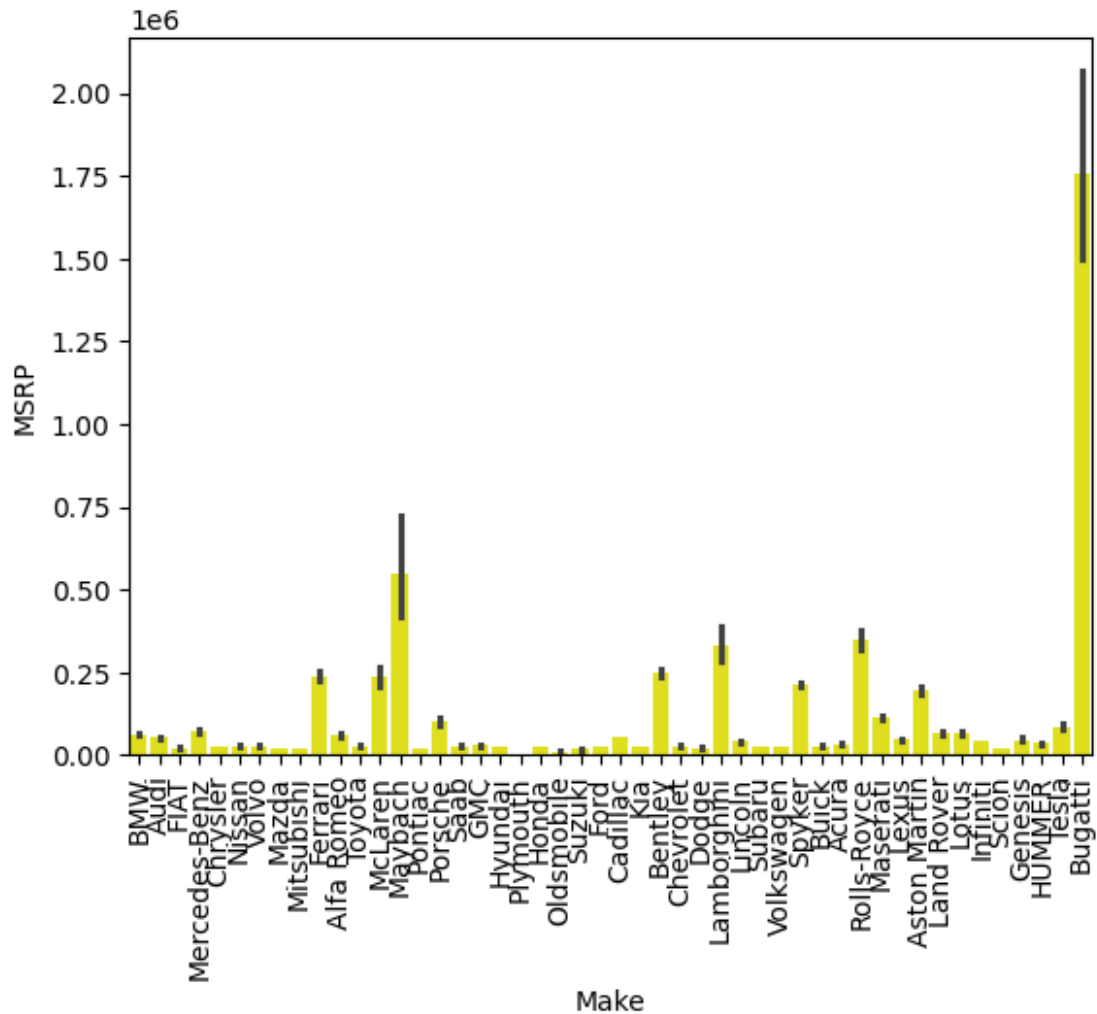


automated manual cars having more prices in the given data set.

```
sns.barplot(data=df,x='Vehicle Size',y='MSRP' ,color='orange')  
plt.xticks(rotation=90)  
plt.show()
```

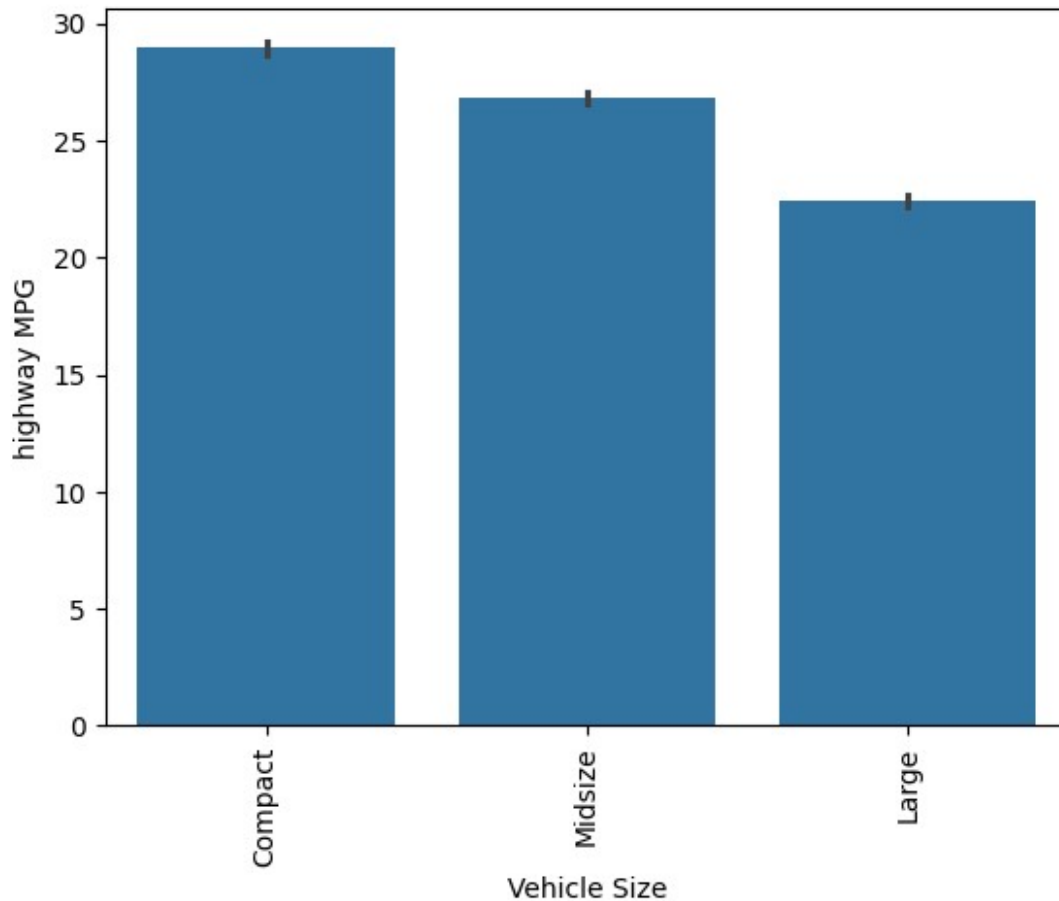


```
sns.barplot(data=df,x='Make',y='MSRP' ,color='yellow')  
plt.xticks(rotation=90)  
plt.show()
```

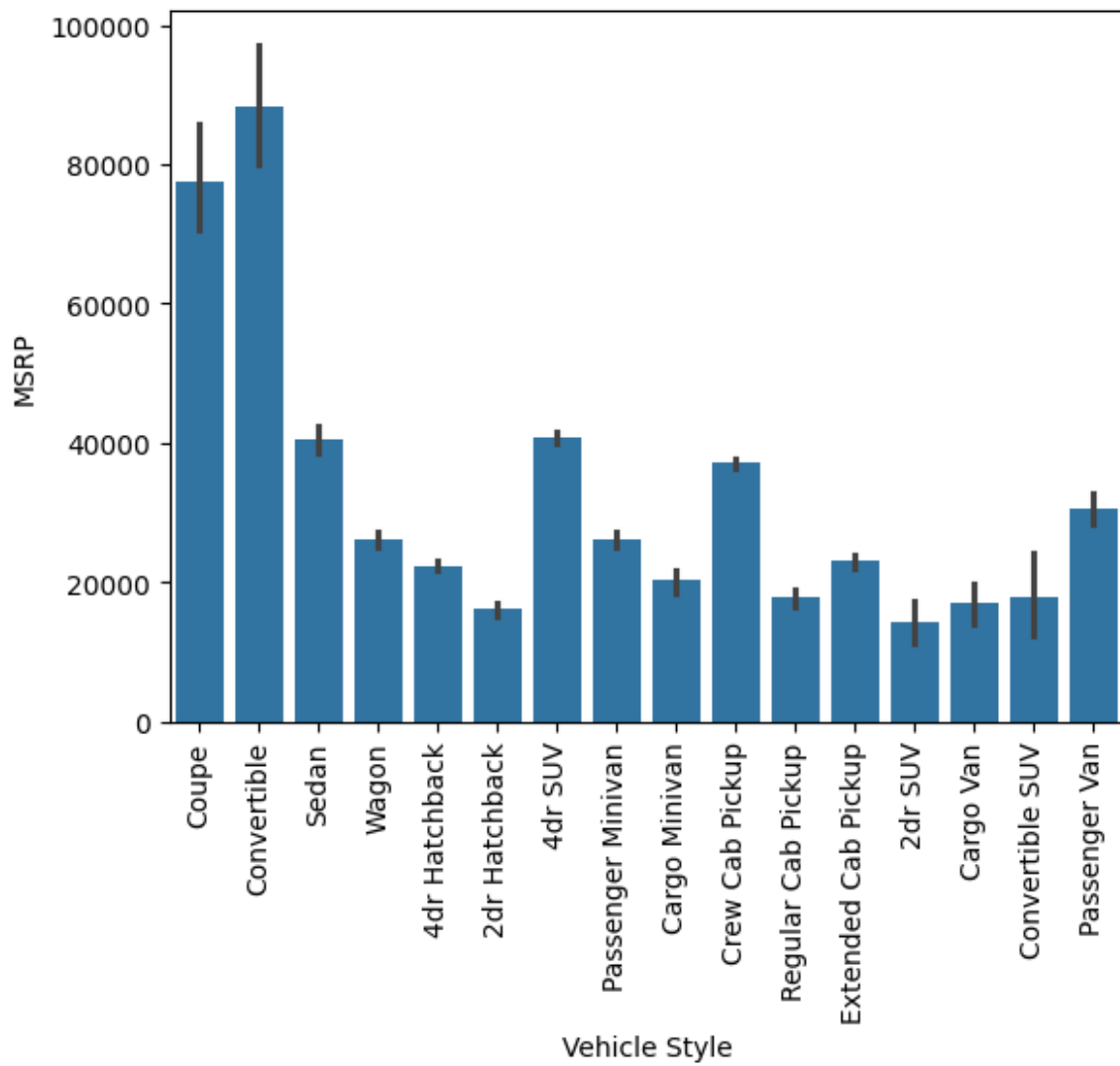
there is relationship btw size of the car and price

```
sns.barplot(data=df,x='Vehicle Size',y='highway MPG')
plt.xticks(rotation=90)
plt.show()
```



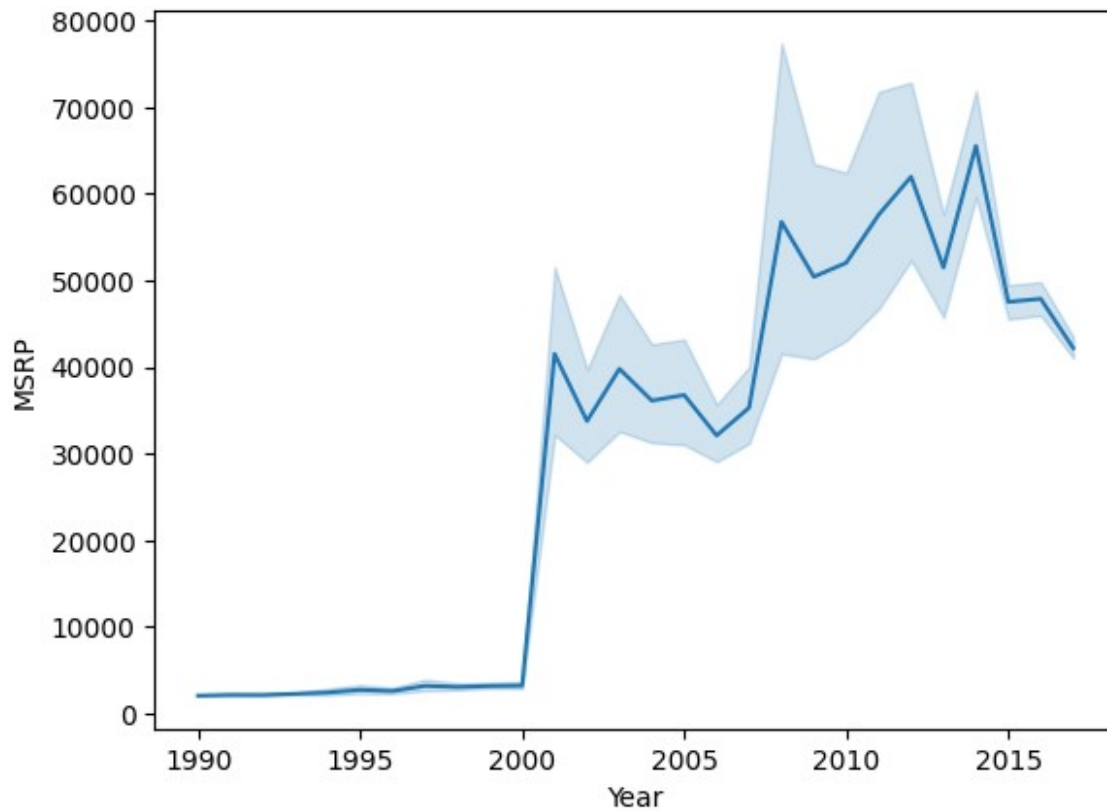
```
## as size of car increasing price increases
df.columns
Index(['Make', 'Model', 'Year', 'Fuel Type', 'Horse Power',
      'Cylinders',
      'Transmission Type', 'Driven_Wheels', 'Number of Doors',
      'Market Category', 'size', 'Vehicle Style', 'highway MPG',
      'city mpg',
      'Popularity', 'MSRP', 'Vehicle '],
      dtype='object')

sns.barplot(data=df, x='Vehicle Style', y='MSRP')
plt.xticks(rotation=90)
plt.show()
```



```
sns.lineplot(data=df,x='Year',y='MSRP')
```

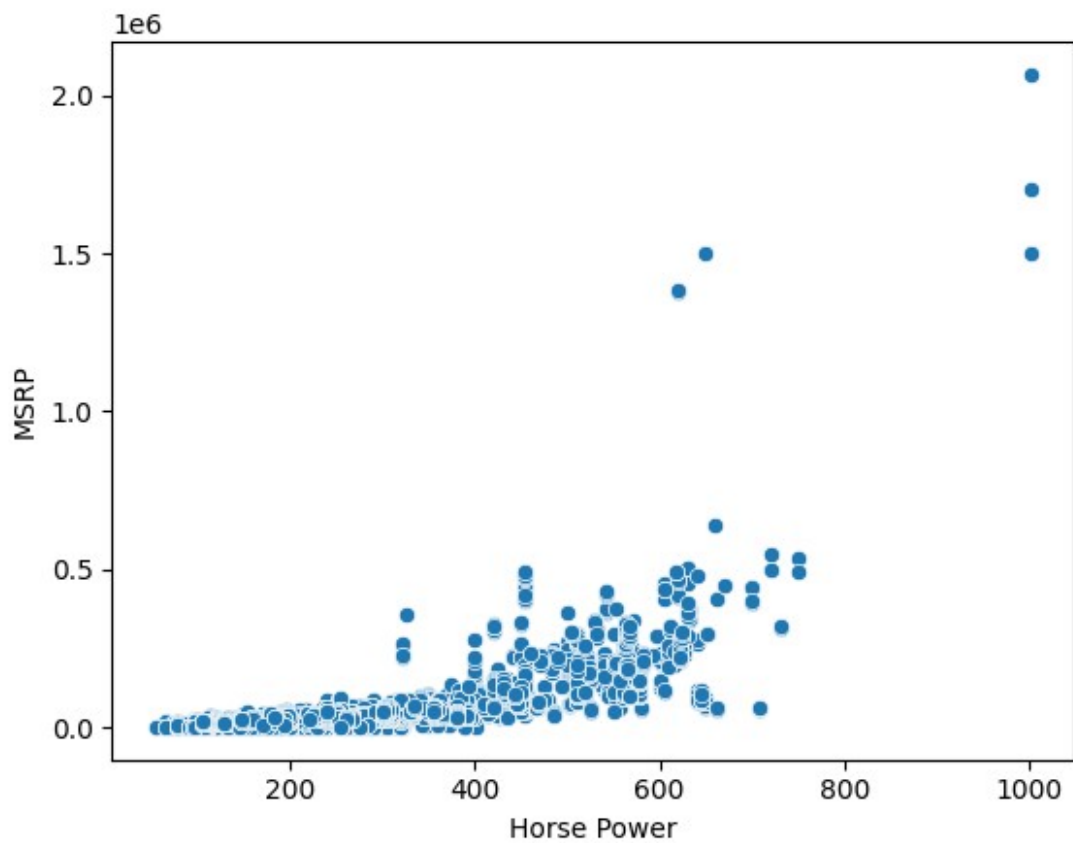
```
<Axes: xlabel='Year', ylabel='MSRP'>
```



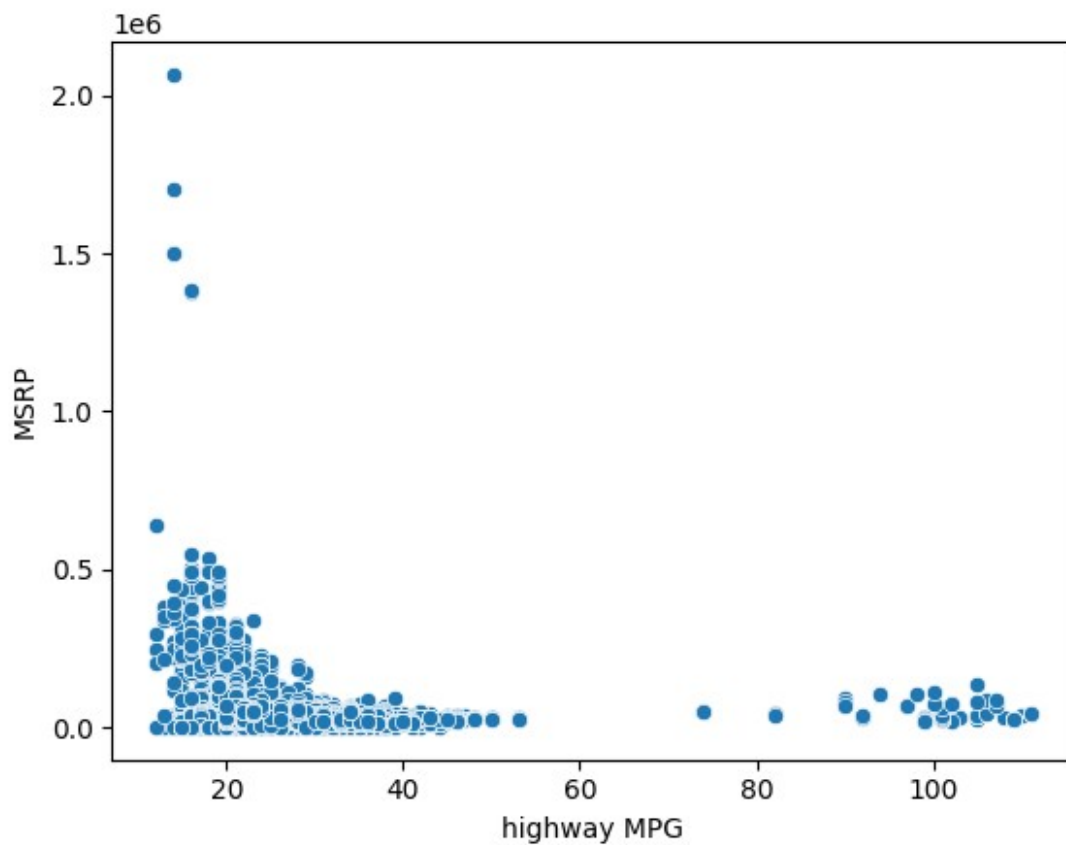
btw 2006 to 2014 car prices are increased and in 2015 there is slight decrement in the prices in given dat set

```
sns.scatterplot(data=df,x='Horse Power',y='MSRP')
```

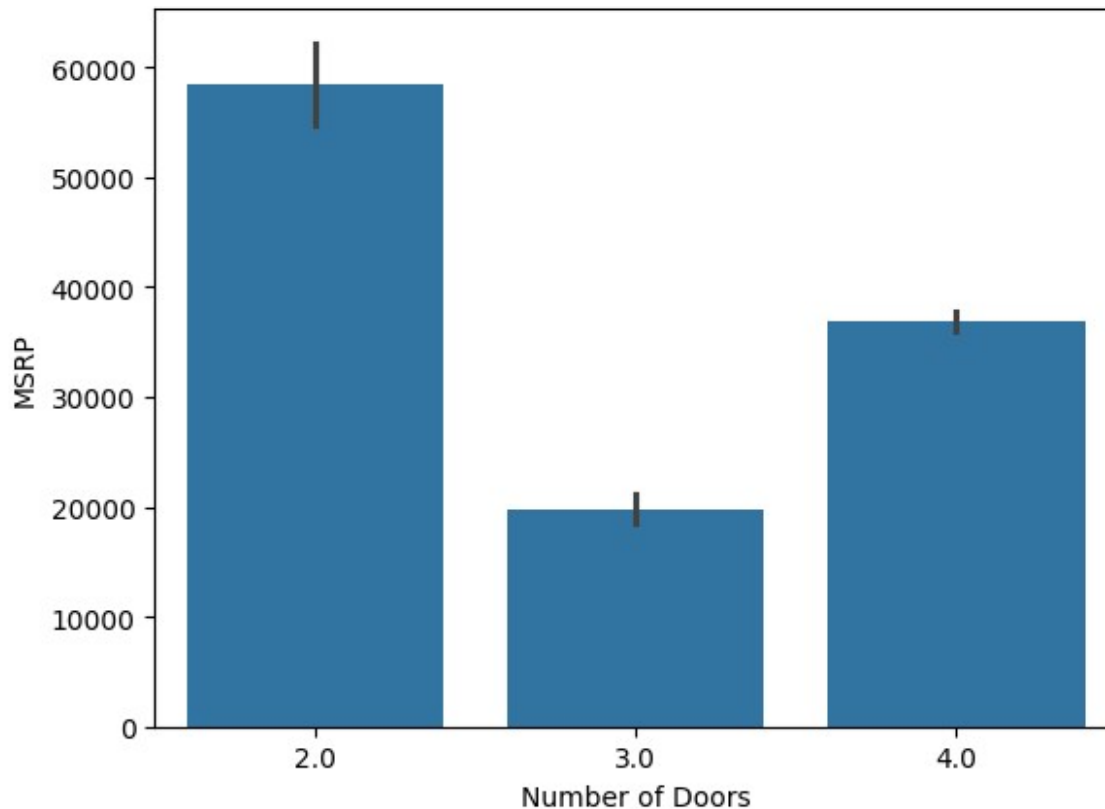
```
<Axes: xlabel='Horse Power', ylabel='MSRP'>
```



```
sns.scatterplot(data=df,x='highway MPG',y='MSRP')  
<Axes: xlabel='highway MPG', ylabel='MSRP'>
```



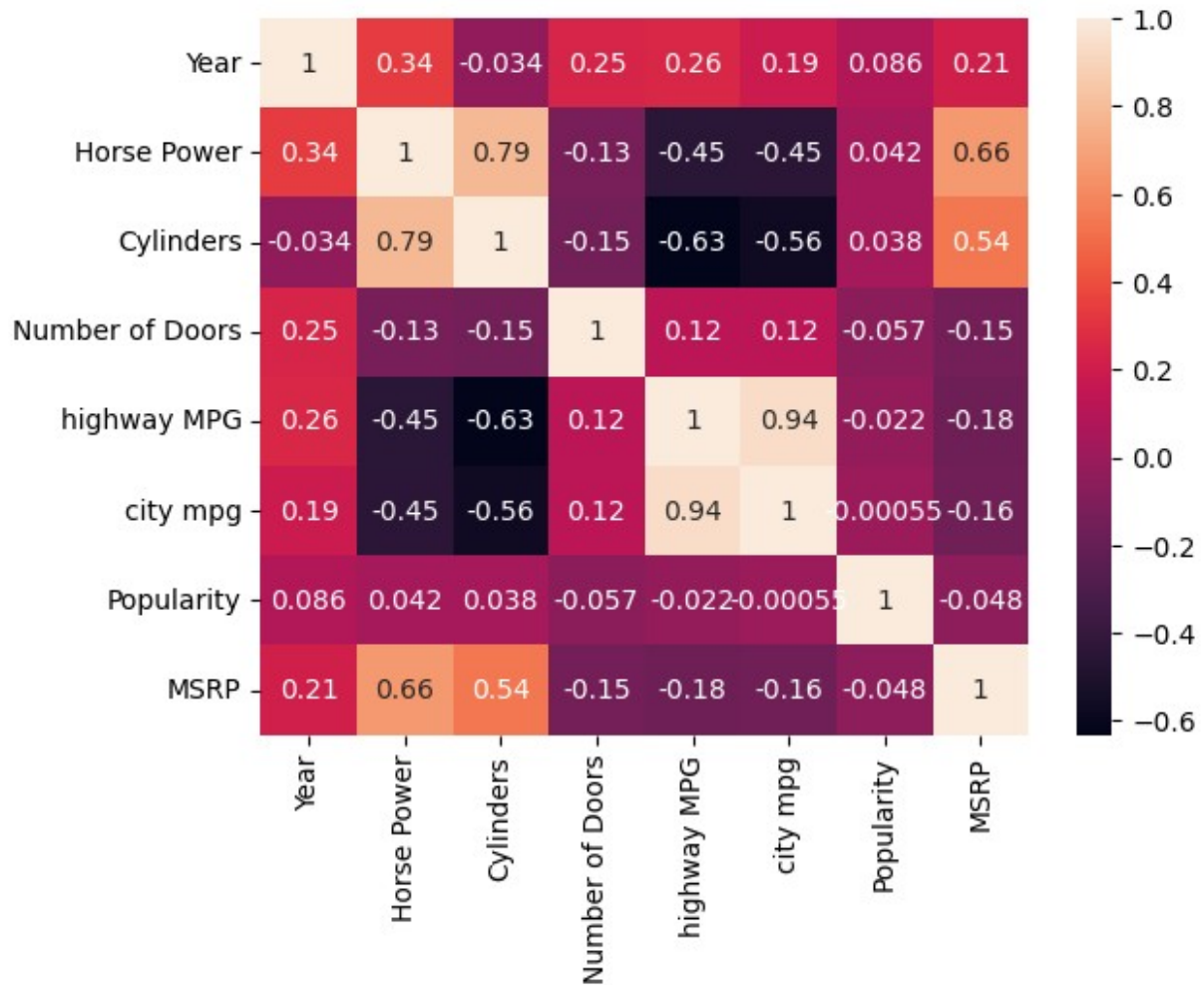
```
sns.barplot(data=df,x='Number of Doors',y='MSRP')  
<Axes: xlabel='Number of Doors', ylabel='MSRP'>
```



```
df.columns
Index(['Make', 'Model', 'Year', 'Fuel Type', 'Horse Power',
      'Cylinders',
      'Transmission Type', 'Driven Wheels', 'Number of Doors',
      'Market Category', 'size', 'Vehicle Style', 'highway MPG',
      'city mpg',
      'Popularity', 'MSRP', 'Vehicle '],
      dtype='object')

sns.heatmap(data=df.corr(numeric_only=True),annot=True)

<Axes: >
```



```
df.columns
```

```
Index(['Make', 'Model', 'Year', 'Engine Fuel Type', 'Engine HP',
      'Engine Cylinders', 'Transmission Type', 'Driven_Wheels',
      'Number of Doors', 'Market Category', 'Vehicle Size', 'Vehicle
      Style',
      'highway MPG', 'city mpg', 'Popularity', 'MSRP'],
      dtype='object')
```